Maximising Exam Marks

Senior Chemistry Book 2

Peter Gribben

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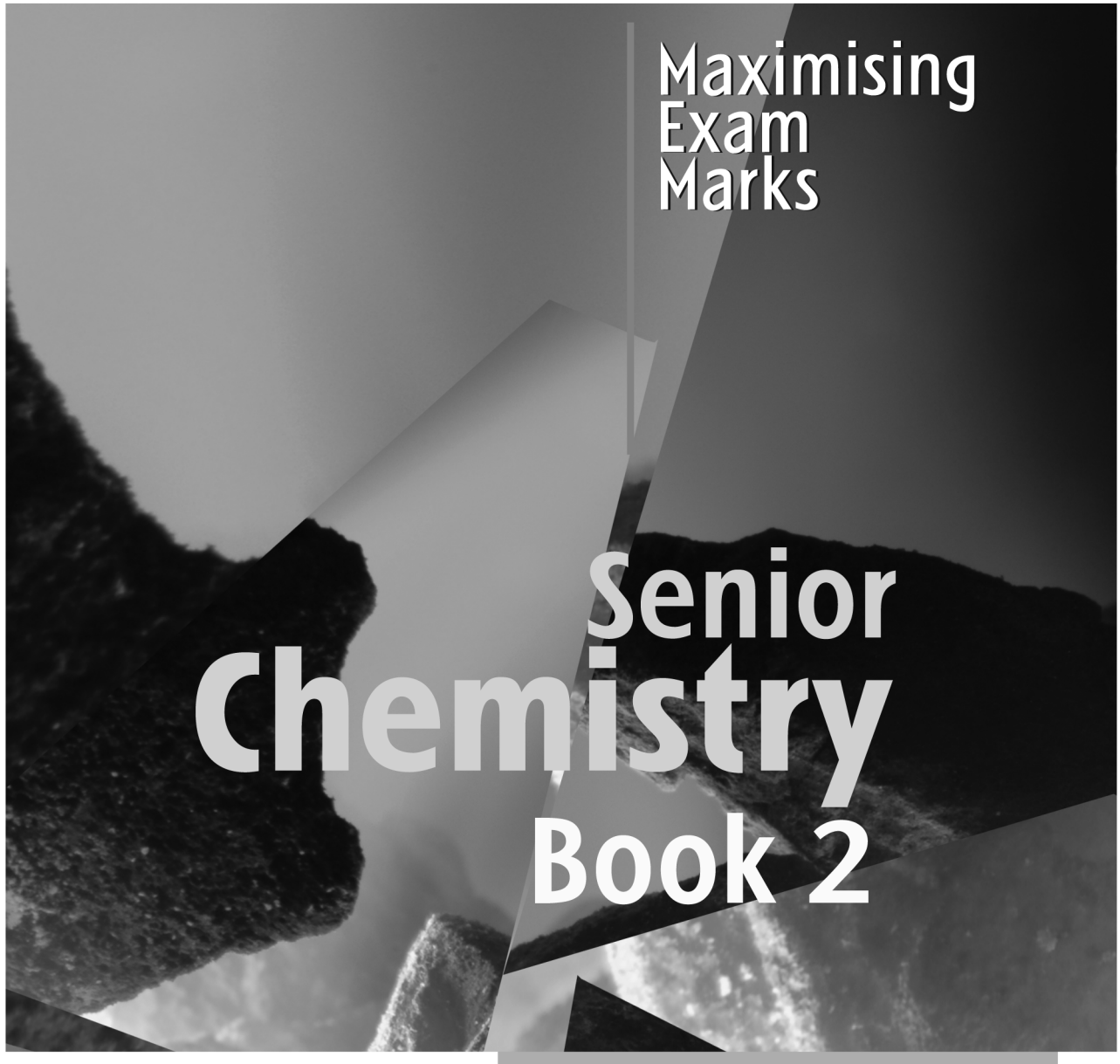

Peter Gribben

This book was commissioned by Jennifer Gough, project management was by Piper Print Media and Gail McManus Graphics designed the cover.

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Printed in Australia by Ligare Book Printer

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An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
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ISBN 978 1 92561 015 4

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INTRODUCTION

Every year, examiners, teachers and assessors in every State produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. This series is designed to assist senior students in areas that they have traditionally found diffficult.

Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

THINGS TO KNOW – Content that students should learn and be able to apply

HINTS – Helping students approach questions correctly

TRICKS – Giving short-cuts or easier ways to attack questions

SLIPS – Providing opportunities to practise working carefully to avoid common errors

TRAPS – Providing the types of contexts in which students make common errors

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1 INTRODUCTION TO ORGANIC CHEMISTRY

The word **organic** is often used to imply that a product is 'natural' or better than other, similar goods. Organic vegetables are produced without the use of pesticides and cost more than other vegetables. Originally a compound was only considered to be organic if it was only produced by an organism. In chemistry, 'organic' has a different meaning – it applies to carbon-based compounds with the exception of oxides and carbonates.

1 *All living things contain carbon compounds, but not all carbon compounds come from living things.* Discuss this statement.

Carbon is only the 14th most abundant element on Earth. However, there over 10 million organic compounds, many times more than in any other class of compounds. There is a huge variety in the size, shape, physical properties and chemical properties of these componds. This is due to a combination of factors. Carbon is in Group IV of the Periodic Table and is tetravalent.

2 What is meant by *tetravalent*?

Figure 1 shows how carbon bonds are arranged in a tetrahedron around the central carbon atom.

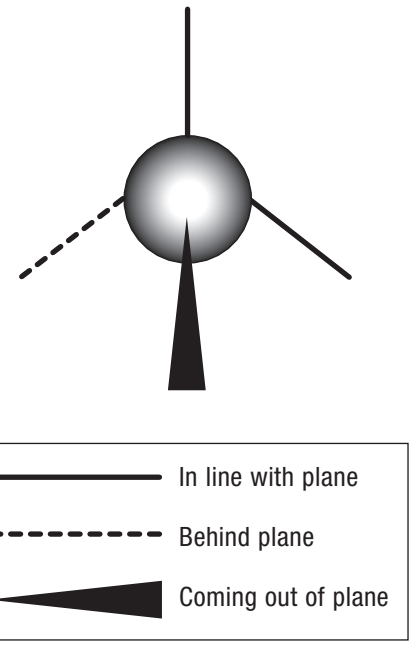


Figure 1

TRAP Contrary to many text book diagrams (necessarily simplified), the bonds in carbon not flat and two-dimensional, but a three-dimensional shape. Each bond represents a pair of shared electrons.

HINT Construct a three-dimensional model to help you visualise the actual shape of molecules.

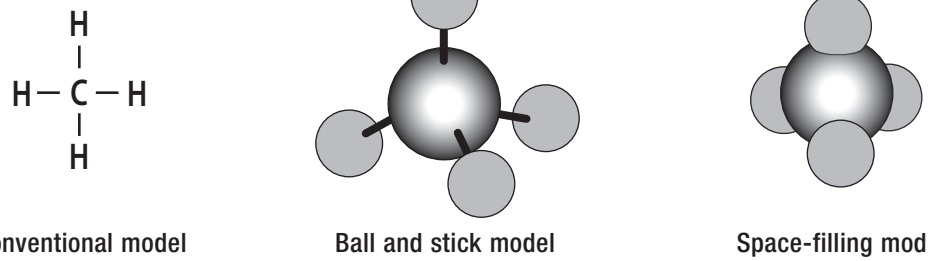
3 Which type of bonding is most likely to be involved with carbon atoms? Explain your answer.

Carbon's ability to form up to four strong covalent bonds at different angles means it can form branching molecules. Carbon also exhibits *concatenation* – the formation of long chain molecules based on a 'backbone' of carbon atoms.

4 Silicon is the other element that can display concatenation. Give a possible reason for this.

HINT Position in Periodic Table

Methane (natural gas) is the simplest organic molecule. Figure 2 shows three ways to represent the structure of methane. The conventional model (also known as the 'structural formula') gives less detail, but is much easier to draw. It gives the untrue impression that methane is a *planar* (flat) molecule.



Conventional model | Ball and stick model | Space-filling model

Figure 2

5 Give the formula for methane.

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Hydrocarbons are compounds containing only the elements hydrogen and carbon. These compounds can be classified into **alkanes**, **alkenes** and **alkynes**.

THINGS TO KNOW

Alkanes are sometimes referred to as 'paraffins' in older text books. Avoid using this term as it can be confused with the liquid fuel also referred to as kerosene.

Alkanes are saturated hydrocarbons, because all the bonds not used to make the chain itself are filled to their capacity (that is, saturated) with hydrogen atoms. Alkanes are a homologous series. That is, they:

- have the same general formula (e.g. C_nH_{2n+2})
- increase in mass as you move up the series
- increase by a constant increment (e.g. CH_2) as you move up the series
- have a gradation in physical properties (e.g. boiling point) through the series
- have very similar chemical properties.

Ethane is the compound following methane in the alkane group.
Figure 3 shows two representations of the ethane molecule.

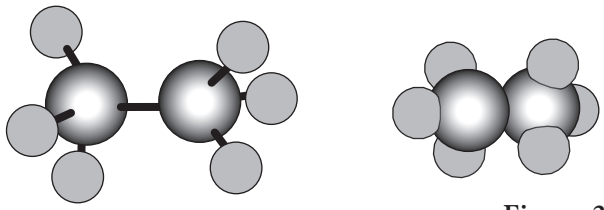

Figure 3

6 Write the formula for ethane and draw its structural formula.

The names, melting points, boiling points and some of the formulae of the first ten alkanes in the homologous series are set out in Table 1.

7 Copy and complete table 1.

HINT

Refer to the points above regarding alkanes.

8 Which of the alkanes listed in table 1 are liquids at room temperature (20°C)?

9 Comment on any trends in the melting points and boiling points of the alkanes shown in table 1.

10 What similarities do you notice in the names of the alkanes?

11 Give the formulae of the next two alkanes after decane.

Table 1: Alkanes - first 10 in the homologous series

Name	Formula	Melting point (°C)	Boiling point (°C)
Methane	CH_4	-183	-162
Ethane	C_2H_6	-172	-89
Propane		-187	-42
Butane	C_4H_{10}	-135	0
Pentane		-136	36
Hexane	C_6H_{14}	-95	69
Heptane		-91	98
Octane		-57	125
Nonane	C_9H_{20}	-51	151
Decane		-30	174

12 Estimate the likely melting and boiling points of the next two alkanes after decane.

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2 NOMENCLATURE

There are many organic molecules so it is important for there to be a system in place to help chemists accurately name and identify these molecules. The International Union of Pure and Applied Chemists (IUPAC) has developed nomenclature for organic compounds, based on their structures. Many organic compounds also have 'trivial' (common) names from the time before the IUPAC system was established.

THINGS TO KNOW Chemical nomenclature is the system of naming compounds.

In the IUPAC system, alkanes (and other organic compounds) are named using a prefix and a suffix. The prefix tells us the number of carbon atoms. The suffix (-ane) tells us that the compound is an alkane; i.e. a hydrocarbon that has only carbon-to-carbon single bonds. The first four prefixes (*meth-*, *eth-*, *prop-* and *but-*) were assigned prior to their structure being known. Prefixes after this are derived from Greek numbers.

Table 1 shows the basis for naming hydrocarbons and other organic compounds.

TRAP Many students have trouble remembering the names of the radicals (chemical groups). A useful mnemonic (memory aid): **M**ax **e**ats **p**eeled **b**ananas, **p**lease **h**and **h**im **o**ne **n**ow **D**ebbie.

Table 1: Hydrocarbon nomenclature

No. carbon atoms in longest chain	Prefix	Formula of radical
1	Meth-	CH_3-
2	Eth-	C_2H_5-
3	Prop-	C_3H_7-
4	But-	C_4H_9-
5	Pent-	C_5H_{11}-
6	Hex-	C_6H_{13}-
7	Hept	C_7H_{15}-
8	Oct-	C_8H_{17}-
9	Non-	C_9H_{19}-
10	Dec-	$C_{10}H_{21}$-

1 Make up your own mnemonic for the prefixes in table 1.

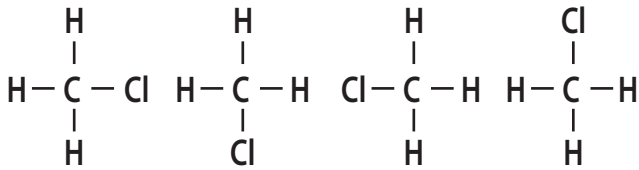
SUBSTITUTED HYDROCARBONS

The hydrogens in an alkane can be replaced by other atoms. When halogens replace hydrogen in alkanes, the resulting compounds are called haloalkanes (IUPAC nomenclature) or alkyl halides.

HINT *Alkyl* means 'derived from an alkane'.

2 What are *halogens*?

Figure 1 shows the structural formula of chloromethane (methyl chloride).

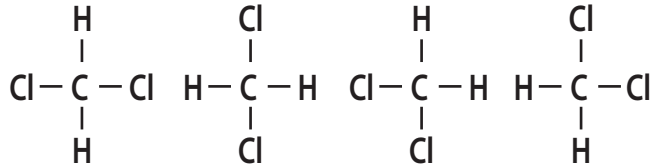


TRICK The structures in figure 1 look different at first glance. If you mentally rotate the structures, you will see they are all equivalent.

Figure 1

3 Give the molecular formula of chloromethane

Figure 2 shows the structure of dichloromethane.



TRICK The structures in figure 2 again may look different, but they are the same. The molecule is three-dimensional (tetrahedral) and the two-dimensional diagram representation used cannot show this.

Figure 2

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Note that each part of figure 2 has a central carbon atom immediately bonded to two hydrogen and two chlorine atoms. Breaking down the name:

- *di* = two
- *chloro* = chlorine
- *methane* = parent alkyl group.

THINGS TO KNOW Learn your terms and derivations.

4 Give the molecular formula of dichloromethane.

5 Draw the structural formula for trichloromethane.

6 Give the molecular formula of trichloromethane.

HINT It is helpful to use three-dimensional models to see how structures that look different in conventional line diagrams are actually the same. Try computer simulations or kits such as Molymod.

Figure 3 shows three representations of chloroethane.

THINGS TO KNOW Carbon-to-carbon single bonds have almost free rotation about their axes. This means that any atoms attached to these carbons are equivalent

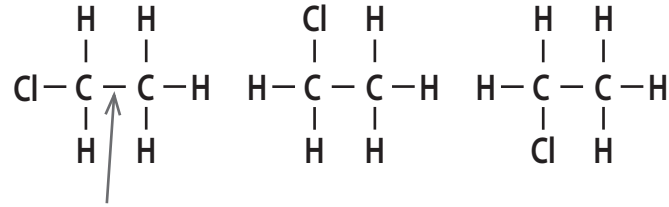

Figure 3

7 Chloroethane can also be called monochloroethane. Give a reason for this.

Figure 4 shows that we can also 'flip' the molecule about the vertical axis and still have the same structure. Another way to visual this is to imagine looking at the molecule from behind.

8 Draw two more representations of chloroethane not already shown on this page.

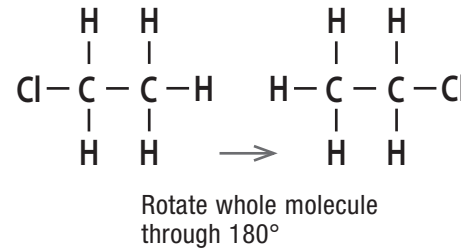

Figure 4

NUMBERING SUBSTITUENTS ON A CARBON CHAIN

When naming an organic compound that has hydrogen atoms replaced by other atoms, for each substituent you need to state:

- their identity
- how many atoms there are (*mono* = one, *di* = two, *tri* = three, *tetra* = four)
- their placement on the carbon chain.

THINGS TO KNOW Know your definitions: a **substituent** is an atom or group of atoms that replaces another atom or group of atoms in a molecule.

9 What other information would be needed when naming an organic compound?

HINT When numbering substituents, go left to right or right to left along the carbon chain, in order to give the lowest number total.

Figure 5 shows different ways to represent 1, 1–dichloroethane.

10 Give the molecular formula for 1, 1–dichloroethane.

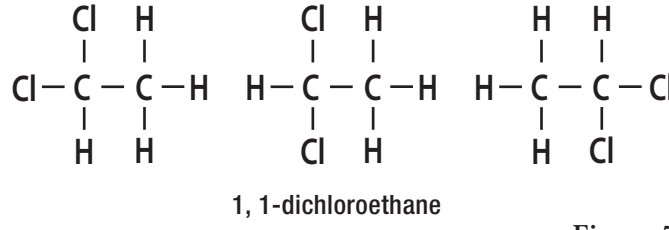

Figure 5

Reading the IUPAC name from right to left can help you understand how a name has been derived.
For example, 1, 1–dichloroethane:

- *ane:* The molecule has been derived from an alkane.
- *eth:* There are two carbon atoms in the chain (from ethane).
- *chloro:* Chlorine has replaced at least one hydrogen atom.
- *di:* Two chlorine atoms are present.
- *1, 1*: Both chlorine atoms are on the first carbon atom of the chain.

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11 Compound (a) in figure 6 is 1, 1-dichoro, 2-bromoethane.

a Give the molecular formula and name for (b).

b Give the molecular formula and name for (c).

c Draw the structure of 1, 2 -dichloro, bromoethane.

d Draw the structure of 1, 1, 2 -trichloro, 2-bromoethane.

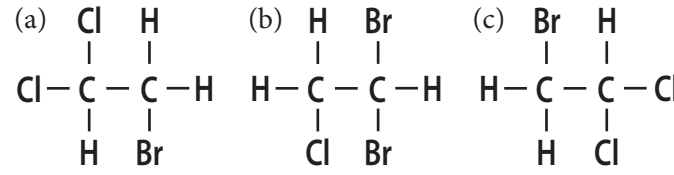

Figure 6

BRANCHED HYDROCARBONS

These hydrocarbons can be regarded as derivatives of straight-chain hydrocarbons, where a hydrogen atom has been replaced by an alkyl radical. Figure 7 shows two ways to represent 2-methylbutane.

The name is derived as follows:

- *ane:* The molecule has been derived from an alkane.
- *but:* The longest carbon chain has four carbon atoms (from butane).
- *methyl:* The methyl group ($-CH_3$) is present.
- *2-:* The methyl group is on the second carbon atom of the chain.

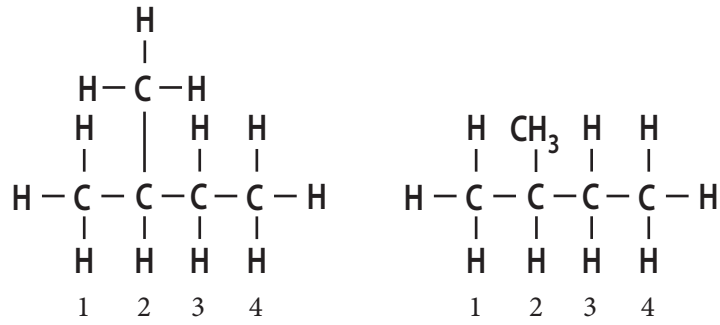

Figure 7

Study figure 8. As you move along the carbon chain in each of these structures, you should see that each chain has five carbons in the main chain and one methyl group attached.

SLIP Note that the bond on carbon 2 is actually the same length as the other carbon-hydrogen bonds.

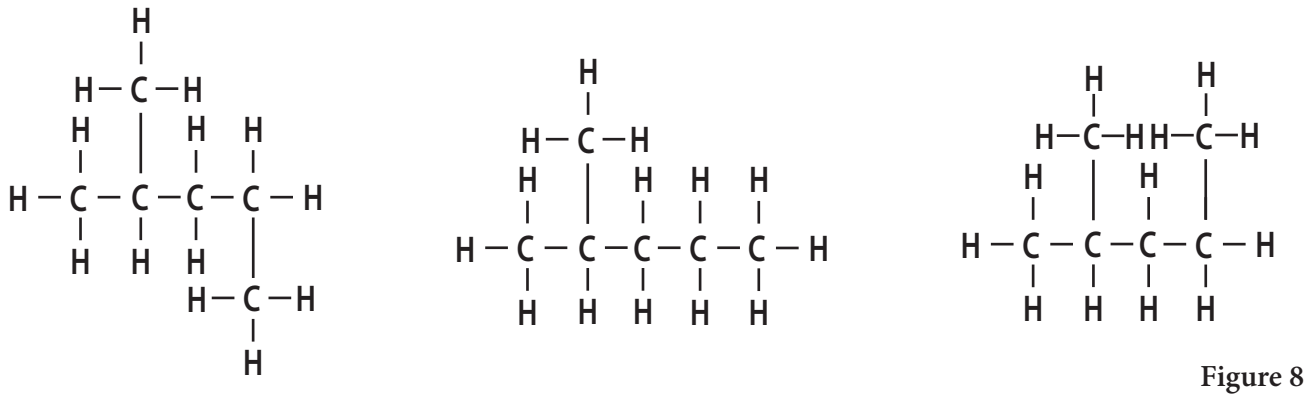

Figure 8

12 Refer to figure 8.

a Give the molecular formula for this molecule.

b Name the molecule.

THINGS TO KNOW Remember that when you work out the name of these types of compounds, as you count along a carbon chain you can go at right angles as well as straight across.

Figure 9 shows three branched hydrocarbons.

13 Refer to figure 9.

a Name compound (a).

b Name compound (b).

c Name compound (c).

14 Draw the structural formula of the compound 3, 3-dimethylhexane, 2-methy, 4-ethyloctane.

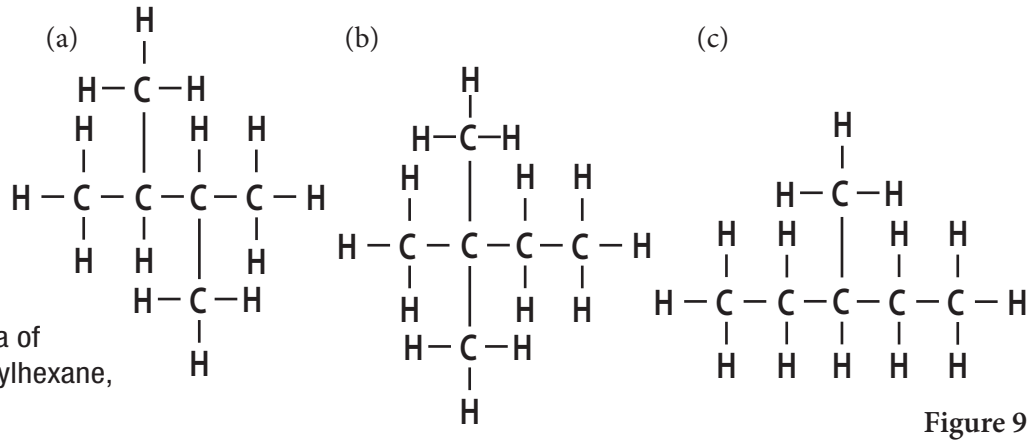

Figure 9

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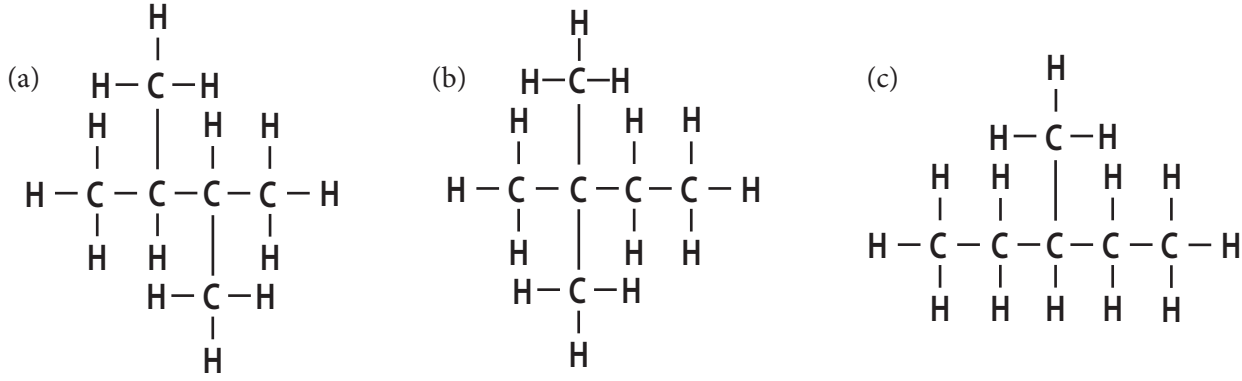
3 ISOMERISM

Isomerism involves compounds that contain the same number and kinds of atoms, but have these atoms arranged in different ways.

Study the molecules shown in figure 1. They are branched-chain hydrocarbons and all have the molecular formula, C_6H_{14}. However, they have different structural formulae.

THINGS TO KNOW

Know your definitions: **Isomers** are compounds that have the same number of atoms of the same elements but have different structural arrangements.



2, 3-dimethylbutane | 2, 2-dimethylbutane | 3-methylpentane

Figure 1

TRAP

Students often confuse **molecular formula** and **structural formula**.

The molecular formula shows the identity and number of atoms present.

The structural formula also shows how these atoms are arranged in relation to each other.

1 Draw and name another isomer with the molecular formula C_6H_{14}.

Study the molecules shown in figure 2. They all have the same molecular formula, $C_3H_4Br_2Cl_2$, but have different structural formulae.

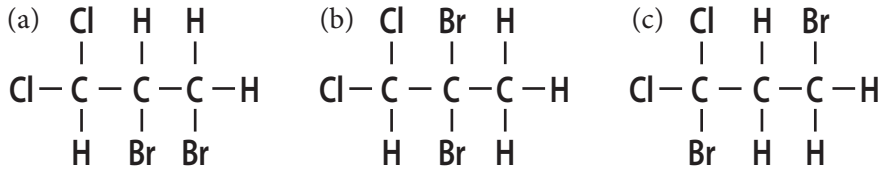


Figure 2

2 Refer to figure 2.

a Give the name for compound (a).

b Give the name for compound (b).

c Give the name for compound (c).

3 Draw and name two more possible isomers with the formula $C_3H_4Br_2Cl_2$.

4 Draw and name two isomers with the formula $C_4H_6Br_2Cl_2$.

HINT

Make sure that you don't have the compound, just drawn in a different way. If in doubt, work out the IUPAC name including numbers.

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4 ALKENES AND ALKYNES

Alkenes and alkynes are unsaturated hydrocarbons. They contain one or more double (alkenes) or triple (alkynes) carbon-to-carbon bonds. Those with more than one double bond or triple bond are referred to as *polyunsaturated*. Alkenes are named in a similar way to the alkanes, using the suffix '*-ene*'.

THINGS TO KNOW An unsaturated hydrocarbon has some carbons atoms able to form other bonds freely.

The simplest alkene is ethene, C_2H_4. Figure 1 shows the three representations of the structure of ethene.

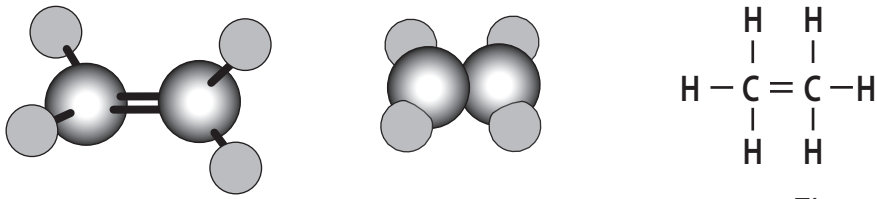

Figure 1

In structural formulae, the carbon-to-carbon double bond is represented by a double line. Note that the atoms immediately connected to the carbon-carbon double bond lie in a plane. The bond angles are bent well away from the ideal value of a tetrahedron. The bond is an area of high electron density and is therefore attractive to electrophiles; i.e. reagents that accept electrons.

THINGS TO KNOW Make sure you have definitions for terms such as ***electrophiles*** and ***reagents***.

1 What does it mean to say a hydrocarbon is *mono-unsaturated*?

The next alkene in the homologous series is propene, C_3H_6. Figure 2 shows its structural formula. Once there are four or more carbon atoms in the chain of an alkene, more than one structure is possible.

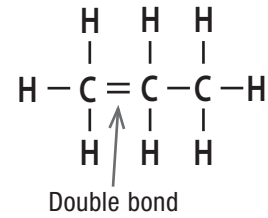

Figure 2

2 What is the name for substances that have the same molecular formula but different structural formulae?

NAMING AN ALKENE

SLIP Some sources may refer to the alkenes as olefins, but this is not a systematic name and so is better avoided.

- It has the suffix *-ene*.
- The longest chain giving the root name must include both of the carbon atoms in the double bond.
- The chain is numbered from the end nearer a carbon involved in the double bond.
- The lower of the two numbers identifying the carbon atoms is used to locate the double bond.
- If more than one double bond is present, the compound is named as a *diene*, *triene*, etc. Each double bond is assigned a number to locate it.

Figure 3 shows two possible structures for C_4H_8.

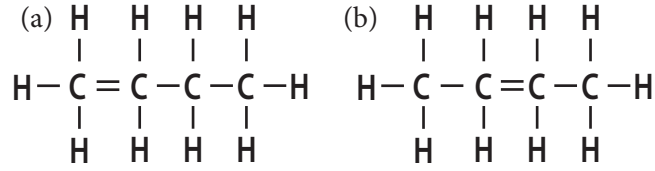

Figure 3

Structure (a) shows the compound but-1-ene. Structure (b) is but-2-ene. If no number is mentioned, it can be assumed that any change to the carbon chain occurs on the first carbon atom; for example, butene is an alternative to but-1-ene.

SLIP Note that using an alternative naming system these two compounds can also be called be 1-butene and 2-butene respectively. Ensure you know which system is used on your course.

TRICK Alkenes with only one carbon-to-carbon double bond have the general formula C_nH_{2n}.

3 Draw and name four molecules with the formula C_5H_{10}.

HINT Branched chains.

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5 ALKYNES

Alkynes are hydrocarbons that contain a carbon-to-carbon triple bond. They are named in a similar way to alkanes and alkenes, but using the suffix '*-yne.*' Figure 1 shows the three representations of the structure of ethyne (C_2H_2) which has the trivial (common) name of acetylene.

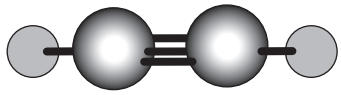
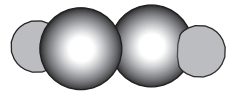

Figure 1

In structural formula, the carbon-to-carbon triple bond is represented by a triple line. Note that the atoms immediately connected to the carbon-carbon triple bond lie in a straight line. In a similar manner to alkenes, the bond angles are under severe strain (180° rather than the preferred 109.5°). The carbon-to-carbon triple bond is an area of even higher electron density than with alkenes. Because of these factors, the carbon-carbon triple bond is extremely reactive.

Ethyne is the most common alkyne. The major use of ethyne is welding – the phrase 'oxy-acetylene' refers to an oxygen and ethyne mixture that gives a very high temperature when burning, up to 2 800°C.

1 Write a word equation and then a balanced formula equation for the combustion of ethyne.

Ethyne can be used to make chloroethene, a raw material used to make PVC, and also 1, 1, 2, 2-tetrachloroethane which is a solvent.

2 Draw the structural formulae for:

a chloroethane

b 1, 1, 2, 2-tetrachloroethane

TRICK Alkynes with only one carbon-to-carbon triple bond have the general formula C_nH_{2n-2}. When naming an alkyne, count carbon atoms from the end of the chain closer to the triple bond.

Figure 2 shows three structures with the molecular formula C_4H_6.

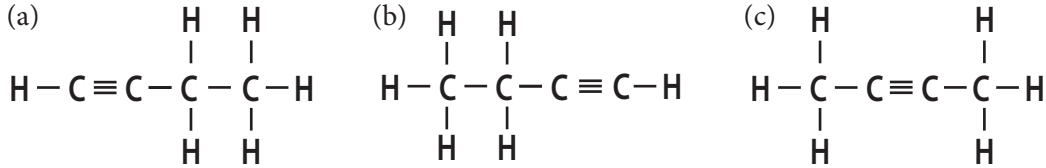

Figure 2

3 Refer to figure 2.

a Name compound (a).

b Name compound (b).

c Name compound (c).

4 Draw the structural formulae for:

a 3-methylbut-1-yne.

b 3, 3-dimethylpen-1-yne.

The chemical reactions of alkynes are very similar to that of alkenes. They both undergo addition reactions and decolourise bromine water. Alkanes do not undergo addition reactions and do not decolourise bromine water.

5 Suggest why the chemical behaviour of the alkanes differs from the alkenes and alkynes.

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6 CYCLIC HYDROCARBONS

CYCLOALKANES

The formula for cycloalkanes is C_nH_{2n}. There are two major groups of cyclic hydrocarbons:

- aliphatic – based on alkanes and alkenes
- aromatic – based on benzene.

Note that some 'aromatic' compounds have strong smells, but many don't – the name is a traditional one. Cycloalkanes have the carbon chain formed into a ring. They are named using the equivalent alkane and with the prefix *cyclo-*. Figure 1 shows cyclopropane, the simplest of the cycloalkanes.

1 Write the molecular formula for cyclopropane.

Cyclopropane is a planar (flat) molecule and the carbon-carbon bond angle is 60° rather than the preferred 109.5°.

2 Describe how the carbon-carbon bond angle will affect the stability of the cyclopropane molecule.

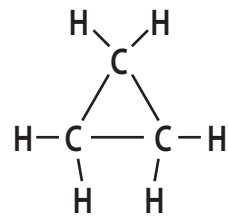
The next two cycloalkanes are cyclobutane (square planar) and cyclopentane (square pentagon)

3 Give the structural and molecular formulae of cyclobutane.

4 Give the structural and molecular formulae of cycloentane.

5 Compare the effect of the carbon-carbon bond angles on the stability of the cyclobutane and cyclopentane molecules.

Structural formula



'Ball and stick'

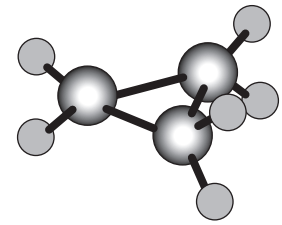
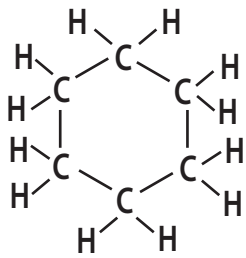


Figure 1

HINT The carbon-carbon bond angle in cyclobutane is 90° and in cyclopentane it is 108°.

Figure 2 shows the arrangement of atoms in cyclohexane. The structural formula implies that the ring in cyclohexane is flat. However, the structure is puckered (twisted) and the 'normal' carbon-to-carbon bond angle is achieved. Hence the ring is not under strain.

Structural formula



'Ball and stick'

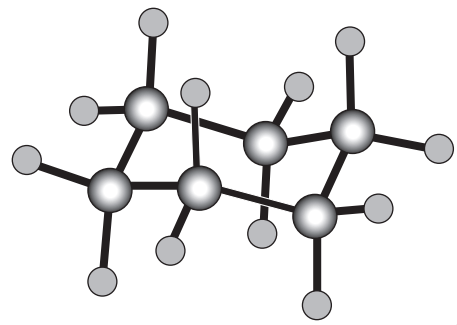


Figure 2

6 Write the molecular formula for cyclohexane.

CYCLOALKENES

Cycloalkenes are named in a similar manner to the cycloalkanes, but with the suffice *-ene*. Like alkenes, cycloalkenes can react by addition to the carbon-carbon double bond and decolourise bromine water. Figure 3 shows the structural formulae of cyclopropene, cyclobutene, cyclopentene and cyclohexene respectively.

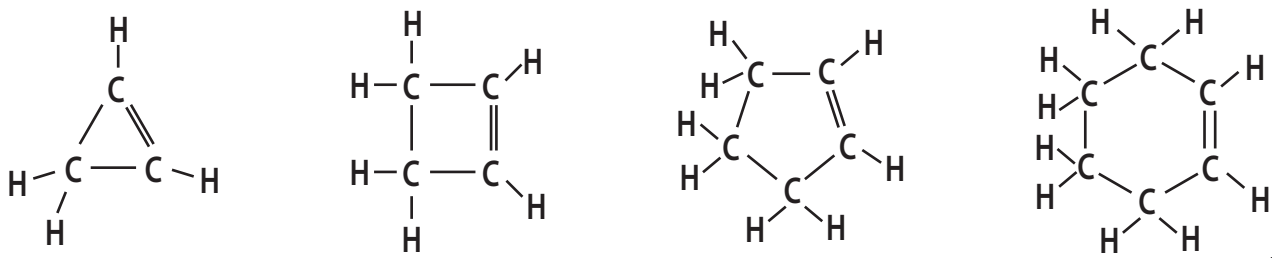


Figure 3

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7 Copy the structural formula of each compound from figure 3 and write the name and molecular formula of the compound below it.

8 Which property of the alkenes could make them suitable as monomers to form polymer chains?

If a cycloalkene has more than one double bond, the positions of these bonds in the carbon ring must be shown. Figure 4 shows the structural formula of cyclohexa 1, 3-diene (1, 3-cyclohexadiene).

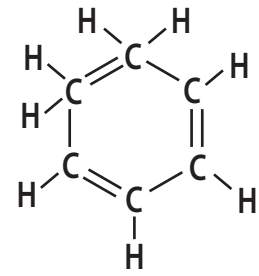

Figure 4

9 Draw the structural formula of cyclohexa 1, 4-diene (1, 4-cyclohexadiene).

10 Are the compounds cyclohexa 1, 3-diene (1, 3-cyclohexadiene) and cyclohexa 1, 4-diene (1, 4-cyclohexadiene) isomers? Explain your answer.

HINT Isomers have the same number and type of atoms, but arranged in a different manner.

AROMATIC HYDROCARBONS

Aromatic hydrocarbons are based on the benzene molecule. The formula of benzene, C_6H_6, suggests unsaturation. It was once suggested that fhe structure of benzene was that shown in figure 5.

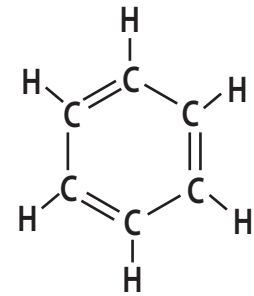

Figure 5

11 What is meant by *unsaturation* when referring to organic compounds?

12 Give the systematic name for the compound with the structure shown in figure 5.

It was found that benzene did not have the structure shown in figure 5. Carbon-carbon single bonds are longer than carbon-carbon double bonds. The bonds in benzene are all the same length, in between the length of single and double bonds, rather than alternating single/double bonds. The benzene molecule tends to react by substitution rather than by addition.

The electrons originally thought to be in the double bonds are delocalised; i.e. spread throughout the ring as shown by the dotted lines in Figure 6.

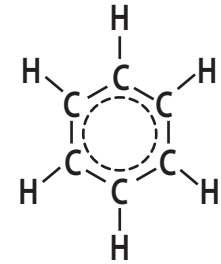

Figure 6

Benzene is not used in school laboratories because it is carcinogenic. Benzene is used as a raw material to make other chemicals. One of these is methylbenzene (toluene). Methylbenzene consist of the benzene ring with one of the hydrogen atoms replaced by a methyl ($-CH_3$) group.

Methylbenzene is a volatile liquid and even in small amounts (100 ppm) causes headaches, dizziness and irritation of the eyes and nose. Long term exposure impairs the central nervous system and causes other chronic effects.

13 What does *carcinogenic* mean?

14 What does *volatile* mean?

15 What do the initials *ppm* represent?

16 What does *chronic* mean?

17 Draw the structural formula of methylbenzene.

18 Write the molecular formula for methylbenzene.

19 **Research:** August Kekulé (1829–96) was a German organic chemist. Find out about his role in determining the structure of benzene.

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7 ALKANOLS (ALCOHOLS)

Alkanols contain the hydroxyl group (-OH) attached to a hydrocarbon framework. Alkanols are named in a simular fashion to alkanes, but have the suffix '*-ol*'. Figure 1 shows how the hydroxyl group is polar, while the hydrocarbon chain is non-polar. This polarity can cause some of the smaller chain alkanols to be miscible with water – which is also polar. Figure 2 shows three ways of representing the simplest alkanol, methanol

TRAP Don't confuse alkanols with inorganic hydroxides. The –OH in alkanols does not form the hydroxide ion when dissolved in water.

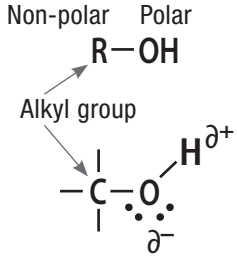


Figure 1

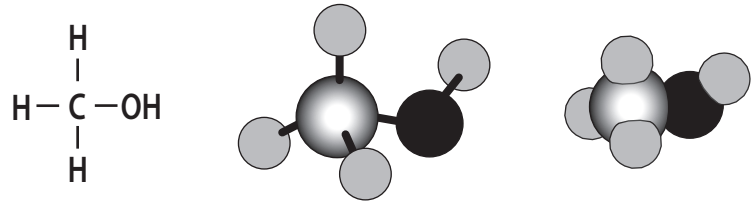
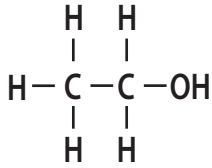
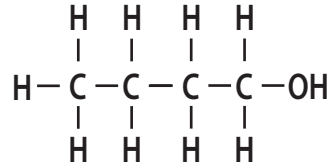


Figure 2

1 Give the molecular formula for methanol.

2 Copy and complete the table to show the four alkanols after methanol in the homologous series.

Name		Propanol		Pentanol
Structural formula				
Molecular formula		C_3H_8O		

3 A text book stated: *Ethanol in some ways can be considered as being an intermediate between ethane and water.* Discuss this statement.

HINT Look at the structure and bonding of all three molecules.

The first four alkanols are miscible with water. As the length of the hydrocarbon chain increases in alkanols, the more the non-polar characteristics tends to predominate over the polar hydroxyl end and the alkanols tend to resemble alkanes in physical characteristics.

Ethanol can be produced by the anaerobic (without air) fermentation of a carbohydrate in water by yeast. An example is glucose ($C_6H_{12}O_6$) solution, producing ethanol and carbon dioxide. Ethanol can be used directly as a fuel or mixed (often 10%) with petrol.

4 Write a balanced formula equation for the fermentation of glucose.

5 Give two uses for ethanol, other than as a fuel.

6 Ethanol burns to form carbon dioxide and water. Write a balanced formula equation for this reaction.

7 **Research:** What is E10 petrol?

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ISOMERS

Alkanols that have three or more carbon atoms can exist as isomers. Figure 3 shows the structural formulae of propanol and propan-2-ol (2-propanol).

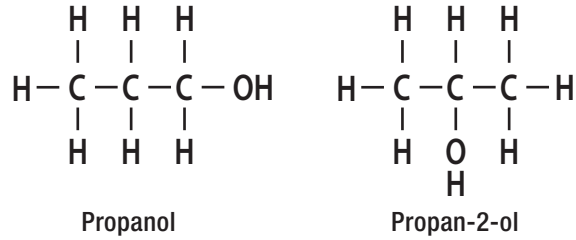

Figure 3

TRICK Note that propanol could also be called propan-1-ol, but if no number is mentioned, it is assumed that the substituent is on the first member of the carbon chain.

Alkanols can have more than hydroxyl group attached. Figure 4 shows the structures and trivial names of a di-ol and a tri-ol. Ethylene glycol is used as an anti-freeze/anti-boil agent in vehicle radiators and glycerol (glycerine) is used in cosmetics, foods and pharmaceuticals.

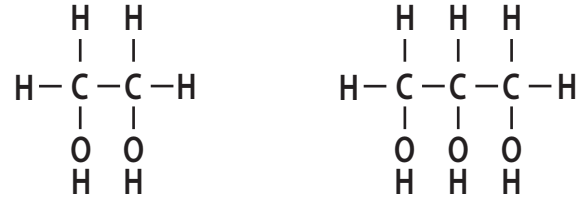

Figure 4

8 Give the molecular formula of propanol.

9 Give the molecular formula of propan-2-ol.

10 Draw the structural formulae of:

a butan-2-ol **b** pentan-3-ol **c** hexan-3-ol.

11 Give the systematic name for ethylene glycol.

12 Give the systematic name for glycerol.

13 Discuss the effect of the 'extra' hydroxyl groups on the physical properties of diols and triols, compared to alkanols that have only one hydroxyl group.

HINT Consider the polar nature of the hydroxyl group.

Figure 5 shows phenol, which used to be known as carbolic acid and was one of the earliest antiseptics. The large carbon ring is non-polar and dominates the hydroxyl group. Phenol is only slightly soluble in water.

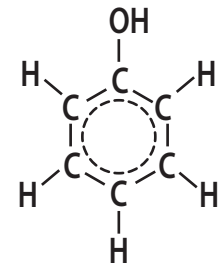

Figure 5

14 What is the name of the type of ring containing six carbon atoms which is shown in figure 5?

Figure 6 shows the structural formula of cholesterol. It is composed of three regions: a hydrocarbon tail, a ring structure region with 4 hydrocarbon rings and a hydroxyl group. Cholesterol functions as a precursor molecule in various metabolic pathways.

15 Copy the diagram shown in figure 6 and label the three regions outlined in the information above.

16 Can cholesterol be considered to be an alkanol? Explain your answer.

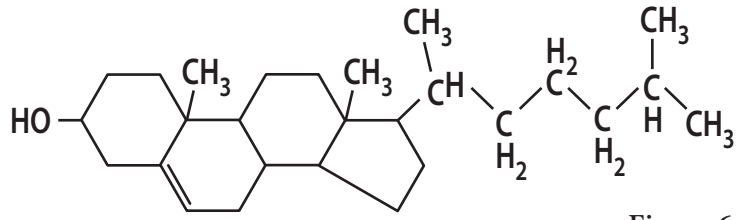

Figure 6

HINT You may need to look up a definition for an alkanol.

17 Define *precursor*.

18 Define *metabolic.*

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8 ORGANIC ACIDS

Organic acids are also called carboxylic acids because they contain the carboxyl (-COOH) functional group. They are weaker than inorganic acids. Their general formula is RCOOH, where R can be hydrogen, an alkyl group or a benzene ring. Figure 1 shows how the functional group is arranged. The alkanoic acids are based on the alkanes and are named in a similar manner but using the suffix '*-oic acid*'. The simplest alkanoic acid is methanoic acid, HCOOH.

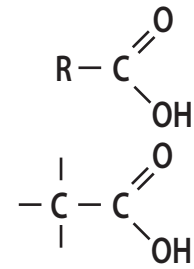

Figure 1

1 Draw the structures of methanoic, ethanoic and propanoic acids.

Methanoic acid is also known as formic acid. This is derived from the Latin term *formica*, meaning 'ant'. Methanoic acid is present in ant bites. Ethanoic acid is also known as acetic acid, from the Latin term *acetum*, meaning 'vinegar'. Household vinegar is about 7% ethanoic acid dissolved in water.

2 Give two uses for vinegar.

When there are two carboxyl groups present, the suffix '*-dioic acid*' is used. Figure 2 shows the structure of an acid with the trivial name oxalic acid. It is a poisonous substance found in rhubarb leaves.

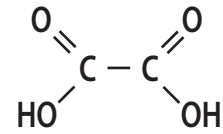

Figure 2

3 Give the formal (IUPAC) name for oxalic acid.

4 Give the molecular formula for oxalic acid.

Figure 3 shows the structure of 2-hydroxypropane-1,2,3-tricarboxylic acid, commonly referred to as citric acid. Citric acid plays a major part in a metabolic process known as the tricarboxylic acid (TCA or 'Kreb's') Cycle.

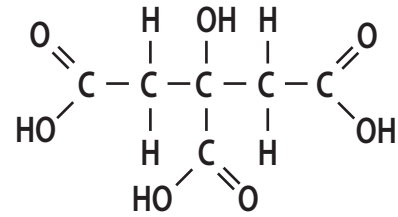

Figure 3

5 Give an everyday use for citric acid.

6 Give the molecular formula for citric acid.

7 Although there are four hydroxyl groups in the molecule, citric acid is described as being a tricarboxylic acid. Explain why.

Benzoic acid (figure 4) is a derivative of benzene and is used as a food preservative.

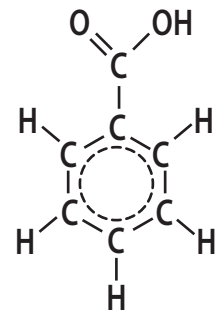

Figure 4

8 Give the molecular formula for benzoic acid.

9 Benzoic is less soluble in water than methanoic, ethanoic and propanoic acids. Give possible reasons for this.

THINGS TO KNOW Learn your terms and derivations.

Like all acids, the carboxylic acids react with bases to form salts. The salts of methanoic acid are called methanoates.

10 Give the names of the salts formed by the following acids.

- **a** Ethanoic (acetic) acid
- **b** Propanoic acid
- **c** Benzoic acid

HINT Many students have trouble remembering the names of the radicals (chemical groups). A useful mnemonic (memory aid) is: **M**ax **e**ats **p**eeled **b**ananas, **p**lease **h**and **h**im **o**ne **n**ow **D**ebbie. (Meth-, Eth-, Prop-, But-, Pent-, Hex-, Hept-, Oct-, Non-, Dec-)

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9 AMINO ACIDS AND PROTEINS

The chemistry of amino acids and proteins looks complex, but a knowledge of the structures of these compounds will help. Amino acids are the monomers that make up proteins. The two major functional groups present in amino acids are the amine and carboxylic (alkanoic) groups.

Figure 1 shows the general structure of amino acids. 'R' represents a side chain, often an alkyl group. The simplest amino acid is glycine, where a single hydrogen takes the place of R.

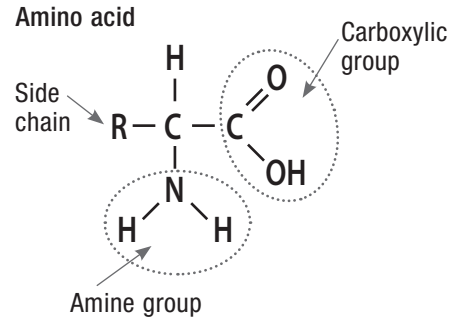

Figure 1

1 Refer to figure 1.

a Give the formula for the amine group.

b Give the formula for the carboxylic group.

2 What is meant by *an alkyl group*?

3 Give the formula for glycine.

The group making up the side chain R in an amino acid be polar or non-polar.

4 What is meant by *polar*?

5 The carboxylic group has acidic properties and the amine group has basic properties. Outline what is meant by the terms *acidic* and *basic* in terms of proton transfer.

HINT Lowry–Brönsted

When amino acids are in solution, they can dissociate (break apart) as shown in figure 2. Notice that the amino acid molecule can simultaneously act as an acid and a base to form a zwitterion.

6 Study figure 2. Using the information above, devise a general description of a zwitterion.

HINT Study the movement of the proton.

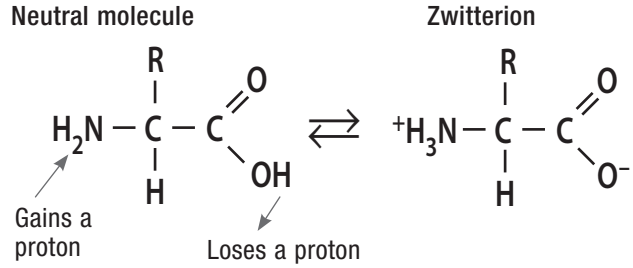

Figure 2

Overall, the molecule is neutral. However, this structure only occurs at a particular pH. In solutions of low pH levels, the carboxylate end of the molecule can gain a proton to form a positive ion.

7 Using figure 2 as a guide, show how a zwitterion can become positively charged when it gains a hydrogen ion.

In solutions of high pH levels, the amine end of the zwitterion can donate a proton to hydroxide ions to form water and a negative ion.

8 Using figure 2 as a guide, show how a zwitterion can become negatively charged when it loses a hydrogen ion.

HINT Hydrogen ions are H^+ ions.

9 *A chemical compound that is electrically neutral but at a specific pH can have positive and negative charges on different atoms.* What does this define?

PROTEINS

The name 'proteins' is derived from the Greek *protos*, meaning 'first'. Proteins are the building blocks of many compounds. Amino acids can link together when an amino group of one molecule bonds with the carboxyl group of another molecule, eliminating a molecule of water in the process. See figure 3.

10 What is the general name for this type of reaction when a molecule of water is 'lost'?

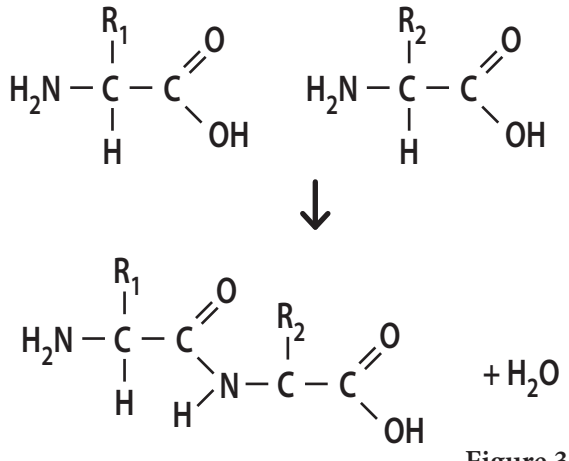

Figure 3

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The linkage between the amino group and carboxyl group is called a peptide bond. (See figure 4.) If two amino acids are combined, it results in a dipeptide. If three amino acids are combined, it results in a tripeptide.

Peptide bond

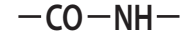

Figure 4

Proteins are polymers where amino acids are the monomers. A typical protein may contain up to 20 different amino acids and be thousands of units in length.

11 Indicate the position of the peptide bond in the molecule shown in figure 3.

12 What would be the general name for a long chain of amino acids joined by peptide bonds?

13 How many peptide bonds would there be in a tripeptide?

14 Identify two factors that might affect the properties of a protein.

PROTEIN BREAKDOWN

Hydrolysis is when a substance is broken down by the action of $H_2O/H^+/OH^-$. Proteins can be broken down by hydrolysis. The usual conditions are warming with moderately concentrated (approx 6 mol L^{-1}) hydrochloric acid

15 If a protein is broken down by hydrolysis, what type of compounds are likely to result?

16 Study figure 5, which shows the hydrolysis of a tripeptide.

HINT Note that the structure is displayed slightly differently.

a What is the missing reactant?

b What is represented by R_1, R_2 and R_3?

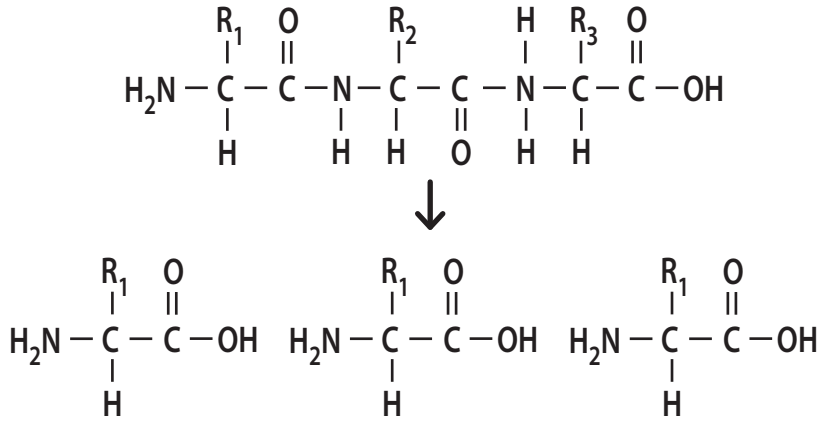

Figure 5

Bonds can be broken at particular places in a protein chain by using specific enzymes.

17 Name a process in the body that uses enzymes to break up protein.

When researching the structure of a protein, scientists work out:

- which amino acids are present
- the proportions of amino acids
- the order in which these acids are arranged.

The protein being investigated is hydrolysed completely into its individual amino acids. The identity and amounts of these acids is found using chromatography and/or electrophoresis. Enzymes are used to break up the original protein at specific peptide links (selective hydrolysis). The resulting fragments are identified and the sequence is worked out.

18 Outline how the sequence of amino acids in a protein could be determined.

The sequence of amino acids is often shown in an abbreviated form, such as Val-His-Leu-Pro, where the three letters stand for the amino acid; i.e. Val is valine, His is histidine, and so on.

TRAP Many students confuse chromatography and electrophoresis.

Chromatography: A process which separates components of a mixture by their different rates of travel through an adsorbing (adhering) medium.

Electrophoresis: a technique which separates large molecules by their different rates of travel through a gel in an electric field.

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19 Suggest why chemists write the sequence of amino acids in an abbreviated form rather than draw the structure.

This sequence is the primary structure of a protein. Because some groups in a protein can be polar, this can affect the shape of the chain.

Amino groups have electron-deficient hydrogen atoms and there are 'lone pairs' (non-bonding electrons) on the oxygen atoms of the carbonyl group. The attraction between these polar groups can cause the chain to fold into a spiral shape as shown in figure 6. (Note that only some of the amino acid links are shown.) This is known as the secondary structure of the protein.

The tertiary structure of a protein is determined by how the spiral folds back on itself in a loop. The structure of a protein can be destroyed (denatured) by heating.

20 Refer to figure 6.

a Explain why there are hydrogen bonds in the protein.

b How does the strength of the hydrogen bonds compare with the covalent bonds present in the chain?

c Show which part of a hydrogen bond is relatively positive (δ^+) and which is relatively negative (δ^-).

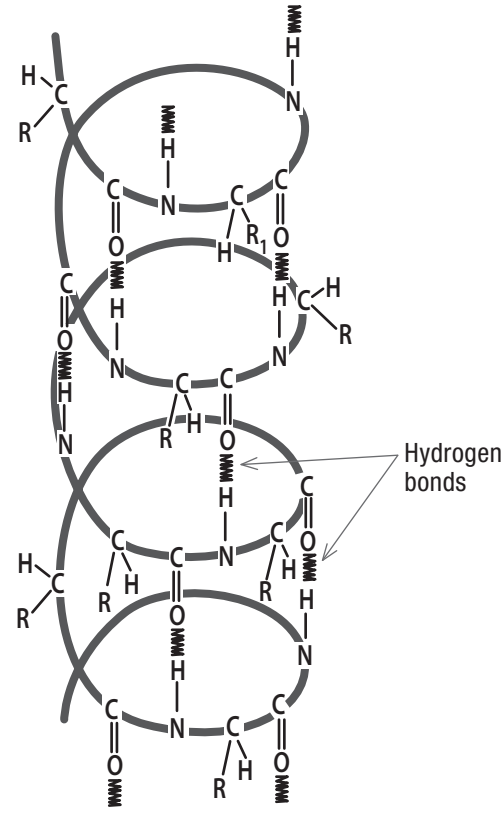

Figure 6

Proteins are important structural materials in animals. Proteins can be classified as either **fibrous** or **globular**.

Fibrous proteins (scleroproteins) are long strings or filaments. They are tough and insoluble in water. They are the components that make up connective tissue in animals, such as the tendons, muscle fibres and bone matrix. Forensic chemists are particularly interested in identifying hair and skin, which are also fibrous proteins. Figure 7 shows the structure of a typical fibrous protein.

21 What is the role of a forensic chemist?

22 What is holding the parallel chains together in figure 7?

HINT Think about bonding.

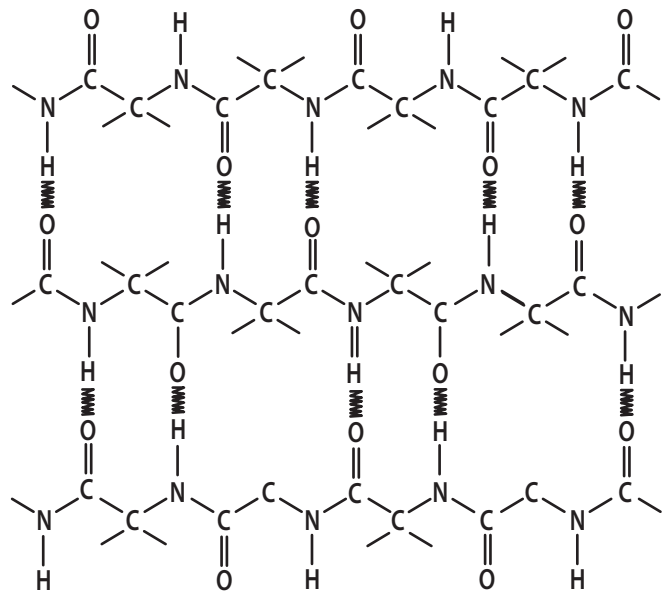

Figure 7

In fibrous proteins, the chains can be twisted into tight spirals. Globular proteins are roughly spherical in shape, more easily denatured than fibrous proteins and more soluble. They have many specialised functions, such as enzymes, antibodies and many hormones. Haemoglobin is a globular protein.

23 What does it mean when a protein is denatured?

24 Identify one role of haemoglobin.

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Globular proteins have polypeptide chains looped and folded back in what looks like a random manner. (See figure 8.) However, the arrangement of these chains is very precise. The specific folds can provide reactive sites for enzymes.

25 Draw up a table and compare fibrous proteins and globular proteins.

TRICK Tables are a good way to present data but can be difficult organise. Use a pencil to set out headings so you can easily amend or remove them.

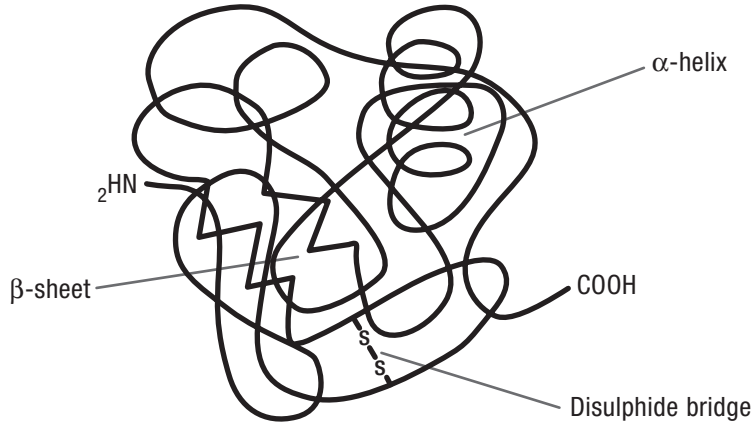

Figure 8

26 *A substance produced by living cells that acts as a catalyst by lowering the activation energy involved in a reaction.* What does this define?

27 All proteins contain carbon, hydrogen, oxygen and which other element?

TESTS FOR AMINO ACIDS AND PROTEINS

Ninhydrin is used to identify the presence of amino acids. Ninhydrin reacts with amino acids to give a purple colour. Ninhydrin spray is also used at crime scenes to show fingerprints, which contain trace amounts of amino acids.

A laboratory technician noted this warning on a spray bottle of ninhydrin solution:
'**Caution**. Ninhydrin solution is toxic by all routes of exposure. Very flammable.'

28 Outline the precautions that would be taken if ninhydrin solution was to be used.

29 Write an equation to show how copper (II) is reduced in Benedict's solution.

HINT Reduction is the gain of electrons.

TRAP Students often confuse Benedict's test with the Biuret test. Both reagents contain copper (II) sulfate and change from a blue colour. However, Benedict's solution is a test for reducing sugars and turns red. The Biuret test is used to identify the presence of proteins.

Biuret solution is an alkaline solution of approximately 1 molar copper (II) sulfate. A deep violet solution is formed when it is mixed with proteins due to the formation of a copper/protein complex.

30 Study the excerpt at the right from a 'Chemical safety in schools' package. Explain what is meant by the terms below.

- **a** Hazardous
- **b** Corrosive
- **c** Toxic
- **d** Ingested
- **e** Fume cupboard

Biuret solution

- Treat as a hazardous substance.
- Use eye and skin protection when handling this corrosive chemical.
- It is toxic if ingested.
- Use a fume cupboard when dealing with quantities of 25 mL or greater.

31 Compare Benedict's test, the Biuret test and Ninhydrin in their roles of identifying compounds.

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10 ADENOSINE TRIPHOSPHATE (ATP)

Most cellular metabolic processes involve adenosine triphosphate (ATP) as an energy source. Figure 1 shows the structure of ATP. Adenine and ribose together form adenosine. The critical part of the molecule is the triphosphate group. Figure 2a shows a simplified version of the ATP molecule. Figure 2b shows another representation of ATP.

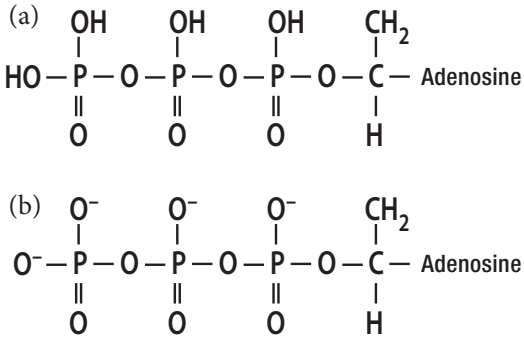


Figure 2

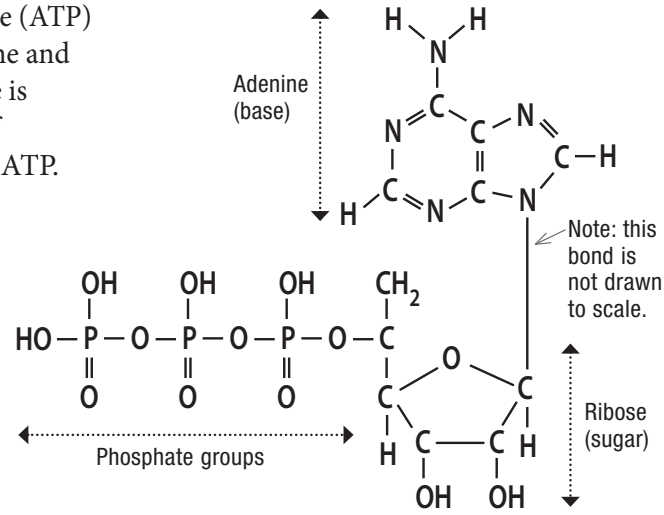


Figure 1

1 What are the differences between the structures shown in figures 2a and 2b?

2 Give the name for the radical shown in figure 3.

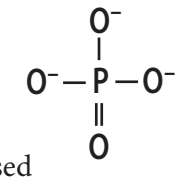
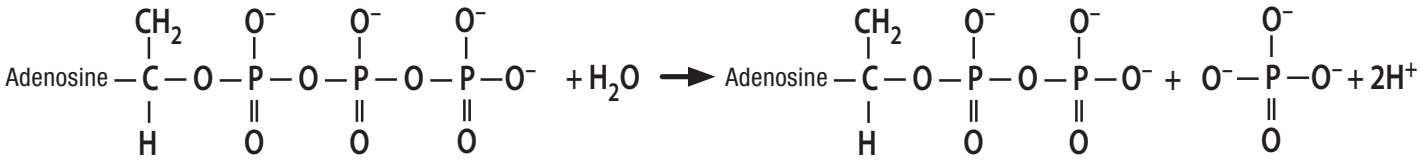


Figure 3

A lot of energy is released when the bonds joining the phosphates are hydrolysed; i.e. broken by the action of water. When the end phosphate group is removed from ATP, around 30 kJ of energy is released per mole. Figure 4 shows this process. (Note that the molecule is shown from a different perspective.)



$$ATP + H_2O \rightarrow ADP + PO_4^{3-} + 2H^+ \qquad \Delta H = -30 \text{ kJ mol}^{-1}$$

Figure 4

3 What is the general term for a reaction that releases energy?

4 Give the full name for ADP.

5 The reaction shown in figure 4 relies on enzymes to occur. Give a definition for *enzyme*.

ATP can be formed from ADP, phosphate ions and hydrogen ions in an endothermic reaction (phosphorylation).

6 Write a balanced equation for the reaction describe above.

TRAP Don't forget ΔH.

The interconversion of ATP and ADP has been likened to that of a 'living battery'. When energy is needed, ATP loses a phosphate group. Energy derived from a process such as respiration of glucose molecules can be used to regenerate ATP from ADP. Cellular respiration has glucose ($C_6H_{12}O_6$) reacting with oxygen forming carbon dioxide and water with the release of 2900 kJ.

7 Write a balanced equation for this reaction of cellular respiration.

TRAP This is an exothermic reaction.

While cellular respiration of glucose is aerobic, ATP is synthesised by mitochondria in an anaerobic process.

8 Define *anaerobic*.

9 What type of energy is present in ATP?

10 Where are mitochondria located?

11 Define *metabolism*.

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11 CALORIMETRY

We use calorimetry to measure the heat changes involved in chemical reactions. The term calorimetry comes from the Latin *calor* meaning 'heat or warmth'. A calorimeter is the apparatus used to carry out calorimetry experiments. A calorimeter can be as simple as a styrofoam coffee cup with a thermometer or a very complex device. Figures 1, 2 and 3 show three common calorimeters used in schools. Calorimetry experiments measure the heat change that occurs for a particular chemical reaction using known quantities of reactants.

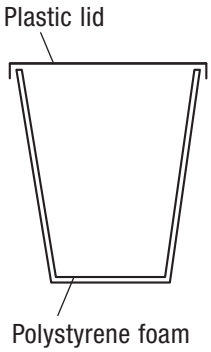

Figure 1

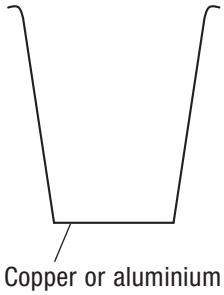

Figure 2

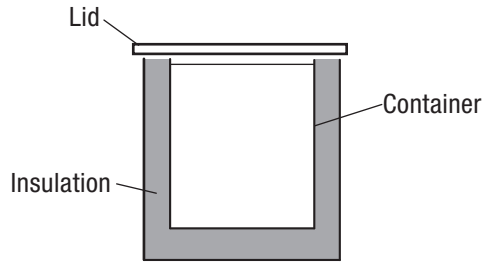

Figure 3

TRAP

Some older textbooks mention calories when dealing with heat experiments. These units are no longer used – they have been replaced by joules (J).

1 Write your own definition for calorimeter.

THINGS TO KNOW

Make sure you have a clear understanding of the concept of exothermic and endothermic reactions.

In exothermic reactions:

- heat is given out
- ΔH is negative
- the temperature of the system rises then falls.

2 Outline the characteristics of endothermic reactions.

When carrying out calorimetry experiments, we measure the temperature change. When you carry out an investigation, ensure that you have measured the maximum temperature change that has occurred. When calculating the enthalpy (heat) change (ΔH) involved in a reaction, you need to measure the mass (m), know the specific heat (Cp) and measure the temperature change (ΔT) of the system. The enthalpy change is calculated using the formula $\Delta H = mCp\Delta T$; i.e. mass of reactants times specific heat times change in temperature.

Study the following carefully.

THINGS TO KNOW

To be successful in calorimetry questions you need to have a good understanding of the equation, terms and units used.

$\Delta H = mC\Delta T$

ΔH = enthalpy change (joules, J)

m = mass (grams, g)

Cp = specific heat (joules per gram per degree, $Jg^{-1}°C^{-1}$) tells us how much energy (J) is required to raise 1 gram of the substance by 1°C (1 K)

ΔT = temperature change (degrees Celsius, °C, or kelvins, K)

3 Give the names and the units involved for the following variables in the equation $\Delta H = mCp\Delta T$.

a ΔH.

b Cp

c ΔT

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Many calorimetry questions deal with acid–base neutralisations and require you to work out the enthalpy of neutralisation per mole. One sequence to follow is:

- work out enthalpy change for the quantities quoted
- work out the number of moles involved (You may have to write a balanced formula equation.)
- calculate the enthalpy change per mole.

WORKED EXAMPLE

A student added 50 mL of 1.0 molar sodium hydroxide to 50 mL of 1.0 molar hydrochloric acid solution in a Styrofoam container. (See figure 4.) Both solutions were initially at 20.1°C. The temperature rose to a maximum of 24.4°C. Assume that the specific heat of the solution is 4.2 $Jg^{-1}°C^{-1}$, that 1.0 mL of solution has a mass of 1.0 g and that no heat is lost to the surroundings.

- **Is this an exothermic or endothermic reaction? Explain your answer.**

  It is exothermic; ΔH is negative, heat must be given out because the temperature initially rises.

- **Calculate the energy change involved in this reaction.**

  Use the equation $\Delta H = mCp\Delta T$.

  $\Delta H = 100\ g \times 4.2\ Jg^{-1} \times 4.3°C = -1806\ J = -1.806\ kJ$

- **Write a balanced equation for the reaction.**

  Sodium hydroxide + hydrochloric acid → sodium chloride + water

  $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$ (Ratio 1:1 base : acid)

- **Calculate the enthalpy of neutralisation per mole for the reaction between hydrochloric acid and sodium hydroxide.**

  Number of moles (n) = (concentration) 1.0 molar × (volume in litres) 0.05 = 0.05 moles

  0.05 moles gives -1.806 kJ, so 1 mole gives -1.806 kJ ÷ 0.05 = -36 kJ

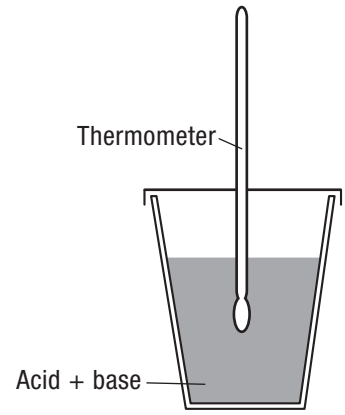

Figure 4

4 In a coffee cup calorimeter, 75.0 mL of 2.0 molar sodium hydroxide and 75.0 mL of 2.0 molar hydrochloric acid were mixed. Both solutions were originally at 24.6°C. After the reaction, the maximum temperature was 34.8°C. Given that the specific heat of the resulting solution is 4.2 $Jg^{-1}°C^{-1}$, calculate the enthalpy of neutralisation (ΔH_{neut}) per mole for the reaction between the base and the acid. Assume that no heat is lost to the surroundings.

TRAP

- Acid + base → salt + water
- Take care with the equation.
- No. of moles = Concentration (moles per litre) × volume (litres)
- Show your working, include units and take care with +/– signs.

5 Explain why a polystyrene foam container is used as a calorimeter in this experiment rather than a copper one.

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12 ENTHALPY OF COMBUSTION

Foods and fuels contain stored energy. Burning them can give an indication of how much energy they have. Combustion involves a substance combining with oxygen and releasing heat and light.

Figure 1 shows how a piece of food (e.g. bread, biscuits, nuts) can be burnt and the energy transferred to a calorimeter. Figure 2 shows a similar process for a liquid fuel such as an alkanol (alcohol).

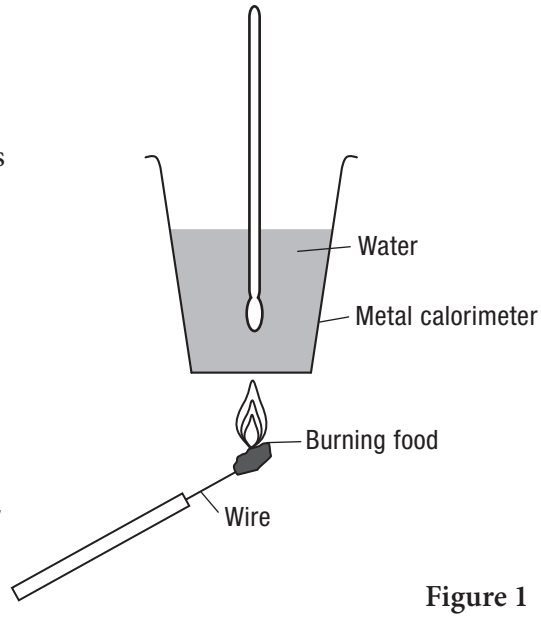

Figure 1

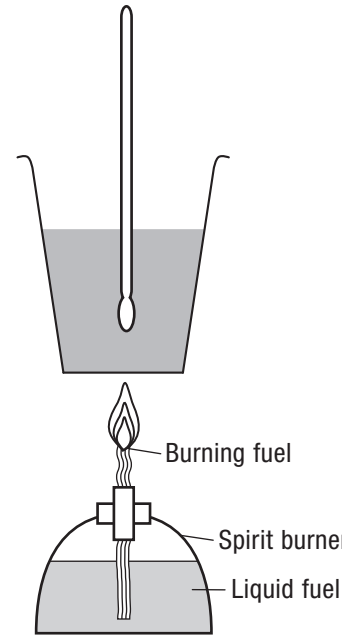

Figure 2

1 Explain why a metal calorimeter rather than a styrofoam one is used in figures 1 and 2.

2 Does all of the heat energy given off by the burning substance enter the calorimeter?
Explain your answer.

When working out enthalpy of combustion, the equation $\Delta H = mC\Delta T$ is used.

Steps:

- Note the mass of substance burned.
- Calculate the energy absorbed by calorimeter.
- Calculate the energy absorbed by contents
- Calculate the energy per gram of substance.
- Calculate the energy per mole of substance.

WORKED EXAMPLE

A student burned a piece of walnut weighing 1.2 g using apparatus as shown in figure 1. The calorimeter weighed 32.3 g, and was made of copper with a specific heat of 0.39 $Jg^{-1}°C^{-1}$. The calorimeter contained 100 mL of water (specific heat 4.2 $Jg^{-1}°C^{-1}$) which went from 20.6°C to 49.2°C.

Calculate the heat of combustion of walnut per gram, assuming that no heat was lost to the surroundings.

- Use the equation $\Delta H = mC\Delta T$.
- Calculate ΔT: 49.2°C – 20.6°C = 28.6°C
- For the calorimeter: ΔH = 32.3 g × 0.39 Jg^{-1} × 28.6°C = 360.3 J
- For the water: ΔH = 100 g × 4.2 Jg^{-1} × 28.6°C = 12 012 J
- Calculatethe total: ΔH = 12 372.3 J, exothermic so -12 372.3 J
- Calculate ΔH per gram: -12 372.3 J ÷ 1.2 = −10 310.2 J = −10.31 kJg^{-1}

3 A student burned a piece of almond weighing 1.1 g using apparatus as shown in figure 5. The calorimeter weighed 34.7 g, and was made of copper with a specific heat of 0.39 $Jg^{-1}°C^{-1}$. The calorimeter contained 100 mL of water (specific heat 4.2 $Jg^{-1}°C^{-1}$) which went from 21.8°C to 43.5°C.

Calculate the almond's energy content per gram, assuming that no heat was lost to the surroundings.

TRAP

When using a metal calorimeter, you have to make allowance for the heat absorbed by the metal part of calorimeter itself as well as that taken in by water.

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BURNING A LIQUID FUEL

Alkanols are commonly used as liquid fuels. The procedure for calculating the molar enthalpy of a liquid fuel usually involves a spirit burner and metal calorimeter. (See Figure 2.) A known mass of fuel is burned; in the process warming the calorimeter and its contents. Molar enthalpy of combustion (ΔH_{comb}) is the heat (J) generated when one mole of a substance is burned completely.

A typical procedure for calculating the molar enthalpy of combustion is:

- weigh spirit burner and contents
- weigh calorimeter
- add water, reweigh calorimeter
- calculate mass of water present
- measure temperature of water
- light the wick of spirit burner and place under calorimeter
- allow fuel to burn until an appreciable temperature rise has occurred.
- extinguish flame, reweigh spirit burner to calculate mass of fuel burned.
- measure maximum temperature rise.
- use equation, $\Delta H = mC\Delta T$ for both calorimeter and contents to calculate enthalpy change
- calculate energy change per gram of fuel
- calculate energy change per mole of fuel.

WORKED EXAMPLE

A student set up apparatus as shown in figure 2. The spirit burner and fuel (ethanol, C_2H_5OH) weighed 137.8 g before it was lit and 134.9 g after it was extinguished. The calorimeter weighed 34.1 g, and was made of aluminium with a specific heat of 0.9 $Jg^{-1}°C^{-1}$. It contained 400 mL of water (specific heat 4.2 $Jg^{-1}°C^{-1}$) which went from 21.4°C to 52.7°C.

Calculate the heat of combustion of ethanol per mole, assuming that no heat was lost to the surroundings.

- Use the equation $\Delta H = mC\Delta T$.
- Calculate ΔT, 52.7°C – 21.4°C = 31.3°C
- For calorimeter, ΔH = 34.1 g × 0.9 Jg^{-1} × 31.3°C = 960.6 J
- For water, ΔH = 400 g × 4.2 Jg^{-1} × 31.3°C^{-1}= 52 584 J
- Calculate total ΔH = 53 544.6 J, exothermic so -53 544.6 J
- Calculate ΔH per gram. Mass of ethanol burned = 137.8 g – 134.9 g = 2.9 g

  -53 544.6 J ÷ 2.9 = -18 463 J = -18.46 kJg^{-1}
- Calculate ΔH per mole. Ethanol is C_2H_5OH = hence molar mass is 46.08 g.

  Enthalpy change per mole = -18.46 kJ × 46.08 = -850.6 kJ

4 A spirit burner and fuel (propanol, C_3H_7OH) weighed 142.8 g before it was lit and 140.3 g after it was extinguished. The calorimeter weighed 34.1 g, and was made of aluminium with a specific heat of 0.9 $Jg^{-1}°C^{-1}$. It contained 400 mL of water (specific heat 4.2 $Jg^{-1}°C^{-1}$) which went from 21.6°C to 62.4°C. Calculate heat of combustion of propanol per mole, assuming that no heat was lost to the surroundings.

(Assume relative atomic masses are C = 12.00 amu, H = 1.08 amu and O = 16.00 amu.)

TRAP

- Don't forget that the question asks for the heat of combustion per mole.
- Be careful when working out molecular masses.

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13 ENZYMES

Enzymes are a special class of proteins that act as very efficient biological catalysts. They have some differences from inorganic catalysts, in that enzymes:

- are often very specific in the reactions they catalyse
- only work over a narrow temperature range
- only work over a narrow pH range.

Individual enzymes are named by adding '*-ase*' onto the name of the substrate with which they reac; e.g. *urease* controls the decomposition of urea. However some have trivial names, such as pepsin.

1 What is meant by a *trivial name*?

Enzymes have an 'active site'. This is an area with a particular structure and shape within which the specific substrate will 'fit'. Intermolecular forces hold the substrate in a particular position. The substitute-enzyme complex can either decompose or react with another substance.

Figure 1 shows a possible sequence of an enzyme-catalysed reaction. The substrate diffuses to the active site on the enzyme and forms an enzyme-substrate complex. The products then diffuse from the active site.

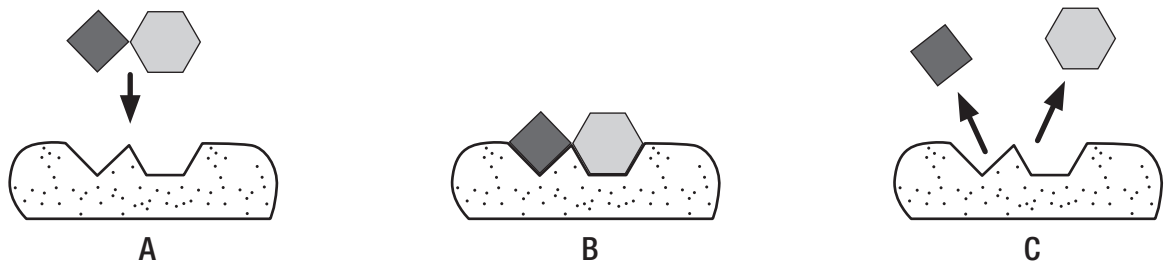

Figure 1

2 Copy figure 1 and identify the active site, the substrate, the enzyme-substrate complex and the products.

3 Discuss the statement: *Enzymes have a binding site that is substrate-specific*.

HINT Look at the shape of the site in Figure 1.

An increase in temperature can speed up enzyme-catalysed reactions, but only to a point. When heated too much, enzymes, because they are proteins, become denatured.

4 What is meant when a protein is *denatured*?

HINT Think about what happens to a protein such as meat when it is heated.

5 Why will an enzyme no longer function once it has been denatured?

HINT Think about the shape of a protein molecule.

Figure 2 shows how the activity of three enzymes varies with pH.

6 Discuss the trends shown in figure 2.

7 Outline the similarities and differences between inorganic catalysts and enzymes.

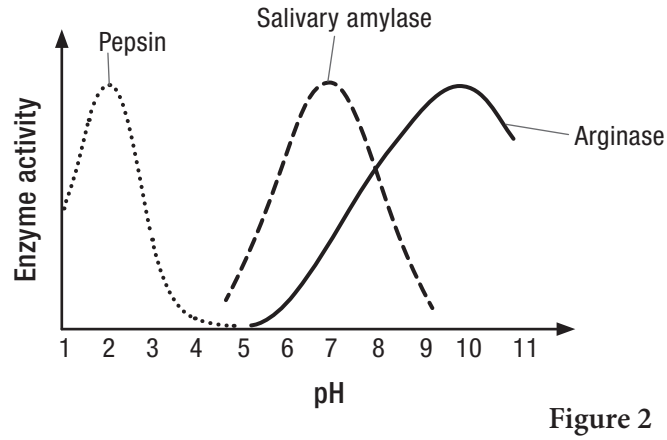

Figure 2

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14 EQUILIBRIUM REACTIONS

When you see the symbol shown in figure 1 in an equation, it signifies that the reaction involved is an equilibrium reaction.

Figure 1

EQUILIBRIUM REACTIONS:

- Convention has the reactants shown on the left and the products on the right.
- The reaction proceeding from left to right is called the forward reaction.
- The reaction proceeding from right to left is called the reverse reaction.
- When the reactants are mixed, the reaction initially goes in the forward direction.
- Once some products are formed there will be a simultaneous reaction going in the reverse direction.
- After a period of time, the rate of forward reaction and the rate of the reverse reaction will be equal. The reaction is then said to be in equilibrium.
- At equilibrium, there is a specific ratio of reactants to products.
- At equilibrium, the mixture usually consists almost entirely of products (position of equilibrium lies to the right) or almost entirely of reactants (position of equilibrium lies to the left), depending on the specific reaction.
- Whether you start with the reactants or products of a particular reaction, the system will eventually reach the same position of equilibrium.

TRAP

The principle behind equilibrium reactions is often misunderstood. A common misconception is that the reaction proceeds from left to right (reactants to products) and then goes right to left (products to reactants) and then left to right and so on – rather like a swing going backwards and forwards.

Another mistaken belief is that the position of equilibrium lies in the centre; that is, that reactions give about 50% of reactants and 50% of products.

1 Draw the equilibrium symbol.

2 Discuss the differences between a reversible reaction and an equilibrium reaction.

3 Explain what is meant by *the position of an equilibrium reaction lies well to the left.*

4 When talking about equilibrium reactions, we often mention 'the system'. Explain what is meant by *system* in this context.

5 A text book stated: *The equilibrium position can be approached from either side of the reaction.* Explain what this means.

TRAP

In order to answer questions on equilibrium reactions, you must have an equation to work from. Many people lose marks by making errors in writing balanced equations. Remember to:

- ensure all formulae are correct.
- show subscripts clearly where needed
- include symbols of state, i.e. (s), (l) (g) and (aq)
- ensure the equation is balanced
- include the equilibrium symbol.

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6 Write balanced formula equations for the following equilibrium reactions.

a Gaseous nitrogen and gaseous hydrogen form the gas ammonia (NH_3)

b Calcium carbonate (a solid) decomposes to form calcium oxide (solid) and carbon dioxide (gas).

c Dinitrogen tetroxide (gas) decomposes to from nitrogen dioxide (gas).

d Methanol (CH_3OH, liquid) reacts with ethanoic acid (CH_3COOH, liquid) to form methyl ethanoate (CH_3COOCH_3, liquid) and water.

TRAP Don't forget to include symbols of state and use the equilibrium symbol.

We can change the ratio of products to reactants when a system has reached equilibrium by altering certain conditions. For example:

- adding or removing a reactant or a product
- changing temperature
- changing pressure/volume (relevant only when gases are involved)
- adding a catalyst to an equilibrium reaction to reduce the time it takes to reach equilibrium.

Because solids take up much less volume for a given mass, their volume is considered negligible compared to gases when studying equilibrium reactions.

7 If you add a catalyst to a reaction that has achieved equilibrium, will it affect the reaction? Explain your answer.

TRAP A catalyst does not affect the position of equilibrium.

Le Châtelier's Principle is extremely important when considering what will happen if conditions are changed in an equilibrium reaction. Many people make the mistake that this is a scientific law or explanation. Not so – it is a **forecast** of what is ***likely*** to happen under certain circumstances. Le Châtelier's principle helps forecast what will happen in an equilibrium system if conditions are changed.

LE CHÂTELIER'S PRINCIPLE

If a system is at equilibrium and it is altered in any way, the system will shift in a way that minimises the effects of the imposed change.

HINT Because textbooks have to summarise Le Châtelier's ideas, you will encounter many versions. Find one that you can understand, remember and apply.

8 Use several different sources to research Le Châtelier's principle. Then write your own version of it.

HINT Ask another student to look at your summary and comment on it.

9 Two students researched Le Châtelier's principle. One student likened it to a game of table tennis with two players batting the ball backwards and forwards. The other student used the example of someone walking down an 'up' escalator at the same speed as it was going up. Discuss how closely these examples match Le Châtelier's principle.

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15 FURTHER EQUILIBRIUM REACTIONS

Many industrial processes involve equilibrium reactions. Manufacturers want these reactions to occur quickly and yield as much product as possible. Knowledge of how equilibrium systems react to change is essential for dealing with these processes.

- If you **add** something to an equilibrium system, Le Châtelier's principle predicts that the system will change in such a way as to **remove** this material.
- If you **remove** materials from an equilibrium system, Le Châtelier's principle predicts that the system will change in such a way as to **replace** this material.

THINGS TO KNOW

Because there are so many possible examination questions on this principle, you need to be able to forecast what will happen as a result of changes to systems.

Iron (Fe^{3+}) and thiocyanate (SCN^-) ions can exist in an equilibrium system as shown:

$$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \rightleftharpoons FeSCN^{2+}_{(aq)}$$

Colourless Red

If you add either $Fe^{3+}_{(aq)}$ or SCN^-, the system will act to remove them, drive the reaction to the right and the mixture will become a darker red. If you add $FeSCN^{2+}$, the system will act to remove it, drive the reaction to the left and the mixture will become a lighter red.

Study this reaction: $SO_{2(g)} + ½O_{2(g)} \rightleftharpoons SO_{3(g)}$

1 Write a word equation for the reaction above.

2 Assume that the reaction above was at equilibrium and in a sealed system. Predict what would happen if you:

a added sulfur dioxide.

b removed sulfur trioxide.

HINT

Revise again a definition of Le Châtelier's principle.

Study this reaction: $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

3 Write a word equation for the reaction.

4 Assume that the reaction above was at equilibrium and in a sealed system. Predict what would happen if you:

a added ammonia.

b removed hydrogen.

Study this reaction: Carbon monoxide gas reacts with water vapour to form methanal (CH_3O) vapour.

5 Write an equation for the reaction.

6 Assume that the reaction above was at equilibrium and in a sealed system. Predict what would happen if you:

a added carbon monoxide.

b removed methanal.

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PRESSURE AND VOLUME

The convention is that we use fixed masses of gases, in sealed containers, at constant temperatures. Standard text books would say something along the lines of:

The volume of a fixed mass of gas at a constant temperature is inversely proportional to pressure.

The pressure of a fixed mass of gas at a constant temperature is inversely proportional to volume. (See figure 1.)

$$\text{Pressure} \propto \frac{1}{\text{Volume}}$$

$$\text{Volume} \propto \frac{1}{\text{Pressure}}$$

Figure 1

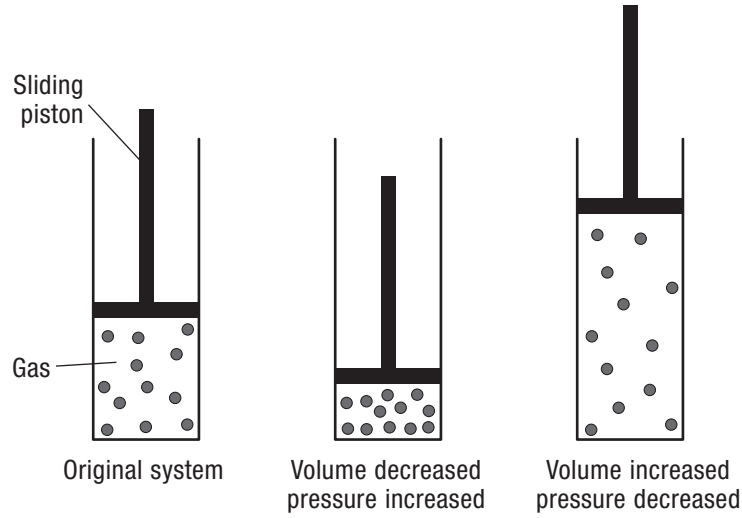

Figure 2

Study figure 2, which shows the relationship between volume and pressure – at a constant temperature.

7 Copy and complete the following passage using these words: *changes, equilibrium, gases, pressure.*

When _____ are involved in an equilibrium system, then _____ in pressure and volume can affect the _____. An increase in _____ will cause the volume to decrease.

The examples below **would** be affected by volume/pressure changes.	The examples below **would not** be affected by volume/pressure changes.
$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ Reactants = 1 mole (1 volume); Products = 2 mole (2 volumes)	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ Reactants = 2 moles (2 volumes) Products = 2 moles (2 volumes)
$SO_{2(g)} + ½O_{2(g)} \rightleftharpoons SO_{3(g)}$ Reactants = 1.5 mole (1.5 volumes) Products = 1 mole (1 volume)	$H_{2(g)} + Cl_{2(g)} \rightleftharpoons 2HCl_{(g)}$ Reactants = 2 moles (2 volumes) Products = 2 moles (2 volumes)
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ Reactants = 4 mole (4 volumes) Products = 2 mole (2 volumes)	

8 In your own words, summarise how you would decide if an equilibrium reaction would be affected by pressure.

9 Decide if the following equilibrium reactions would be affected by a change in pressure. Explain your answers.

a $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

b $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

c $C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$

TRAP Changing the pressure and volume will only affect the equilibrium in reactions that involve gases and where the number of moles of gas differ between reactant and product.

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16 CALCULATING pH

The pH scale describes the concentration of hydrogen ions in a solution. It is logarithmic scale, which means it is based on powers of ten – this is because the concentration of hydrogen ions can vary immensely. Think of how your calculator uses scientific notation for very small and very large numbers.

1 Study table 1. What do you notice about the relationship between hydrogen ion concentration and pH?

Concentration of hydrogen ions (mol. L^{-1})		pH
0.1	10^{-1}	1
0.01	10^{-2}	2
0.001	10^{-3}	3
0.000 1	10^{-4}	4
0.000 01	10^{-5}	5
0.000 001	10^{-6}	6
0.000 000 1	10^{-7}	7
0.000 000 01	10^{-8}	8
0.000 000 001	10^{-9}	9
0.000 000 000 1	10^{-10}	10
0.000 000 000 01	10^{-11}	11
0.000 000 000 001	10^{-12}	12
0.000 000 000 000 1	10^{-13}	13
0.000 000 000 000 01	10^{-14}	14

Table 1

THINGS TO KNOW

A change of one pH unit is equivalent to a tenfold change in hydrogen ion concentration.

CALCULATING pH

pH = $-\log_{10}[H^+]$

[H+] = hydrogen ion concentration in moles per litre

The molarity of a solution can be shown as M or mol.L^{-1}

WORKED EXAMPLE

What is the pH of a 0.2 mol.L^{-1} solution of nitric acid? (Assume that the acid is completely dissociated.)

- pH = $-\log_{10}[H^+]$
- pH = $-\log_{10}0.2$
- Press log function (log) on your calculator
- Enter 0.2
- Press equals (=)
- Press change sign (+/–)

Answer: pH = 0.70

TRAP

Ensure that you are familiar with the logic of your calculator. The sequence may vary from the one shown here.

You may also need to use pH to work out hydrogen ion concentration.

WORKED EXAMPLE

What is the hydrogen ion concentration of a solution of pH 4.5?

- 4.5 = $-\log_{10}[H^+]$
- Press shift/second function on your calculator
- Press log function
- Enter 4.5
- Press change sign (+/–)
- Press equals (=)

Answer: hydrogen ion concentration = 3.2×10^{-5} mol.L^{-1}

HINT

If you don't get this answer, try changing the sequence on your calculator. Make a note of the actual sequence required.

2 Calculate the pH of the following solutions.

a 0.02 mol. L^{-1} hydrochloric acid

b 0.15 mol. L^{-1} nitric acid

c 0.01 mol. L^{-1} sulfuric acid

TRAP

When dealing with sulfuric acid (H_2SO_4), don't forget that for every mole of acid, two moles of hydrogen ions are produced.

3 Calculate the hydrogen ion concentration for a solution that has a pH of:

a 4.1. **b** 13.0.

c 7.45. **d** 1.1.

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17 BUFFER SOLUTIONS

If you add acids or bases – even in small amounts – to water and most solutions, the pH can change considerably. Buffer solutions are less susceptible to changes in pH when small amounts of acids (H^+) or bases (OH^-) are added. Buffer solutions are not necessarily neutral; they can be formulated to a wide variety of pHs. They can be used to:

- calibrate pH meters
- maintain an optimum pH for industrial processes e.g. dyeing cloth
- keep pH within a certain range for enzymes e.g. involved in body processes.

TRAP Buffer solutions only resist changes in pH when small amounts of acids or bases are added, they can't resist changes effectively if large amounts are added.

An acidic buffer solution contains a weak acid and its salt, often the sodium salt. Basic buffer solutions contain a weak base and its salt. Buffer solutions are examples of equilibrium systems and a knowledge of Le Châtelier's principle can help you understand how these systems work.

1 Name two weak acids.

2 Name two weak bases.

3 Outline Le Châtelier's principle.

Study the equation below showing the dissociation of ethanoic acid.

$$CH_3COOH_{(l)} + H_2O_{(l)} \rightleftharpoons CH_3COO^-_{(aq)} + H_3O^+_{(aq)}$$

HINT Remember that weak acids only partially dissociate when dissolved in water.

4 Would the position of equilibrium of this system lie to the right (products) or to the left (reactants)? Explain your answer.

A buffer solution involving ethanoic acid would have a salt of this acid added, e.g. sodium ethanoate, CH_3COONa. In aqueous solution this would dissociate as shown:

$$CH_3COONa_{(s)} + H_2O_{(l)} \rightleftharpoons CH_3COO^-_{(aq)} + Na^+_{(aq)}$$

5 Sodium ethanoate is a strong electrolyte. Explain how the dissociation would be affected.

When the equilibrium system is described for the buffer solution made up of ethanoic acid and sodium ethanoate, the sodium ion is usually omitted because it is regarded as a 'spectator ion'.

6 Explain what a *spectator ion* is.

HINT Think about the normal use of the word 'spectator'.

In this mixture, the acid is almost completely undissociated and the salt almost completely dissociated as shown:

$$\underset{\text{From acid}}{CH_3COOH_{(l)}} + H_2O_{(l)} \rightleftharpoons \underset{\text{Mainly from salt}}{CH_3COO^-_{(aq)}} + H_3O^+_{(aq)}$$

Buffer systems are equilibrium systems involving conjugate acid–base pairs**.** An acid (e.g. Acid 1) and its conjugate base (Base 1) only differ by H^+.

$$\underset{\text{Acid 1}}{HA_{(aq)}} + H_2O_{(l)} \rightleftharpoons \underset{\text{Base 1}}{A^-_{(aq)}} + \underset{\text{Acid 2}}{H_3O^+_{(aq)}}$$
Base 2

HINT After writing out the equilibrium system, study the example shown above carefully.

7 Write out the equilibrium system involving the mixture of ethanoic acid and sodium ethanoate in solution. Identify the conjugate acid–base pairs present.

But how do buffer solutions work? They have components that remove hydrogen ions or any hydroxide ions that are added. What will happen to this system if hydrogen ions are added? Le Châtelier's principle would imply that these hydrogen ions will combine with the ethanoate ions already present to form ethanoic acid.

8 Write a balanced formula equation for the reaction described above.

9 In which direction does this drive the overall equilibrium reaction?

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What will happen to this system if hydroxide ions are added? Le Châtelier's principle would imply that these hydroxide ions will combine with the hydrogen ions already present to form water.

10 Write a balanced formula equation for the reaction described above.

11 In which direction does this drive the overall equilibrium reaction?

Another acidic buffer solution is a solution of carbonic acid (H_2CO_3) and hydrogencarbonate (bicarbonate, HCO_3^-) ions.

12 Suggest a compound that would dissociate almost completely to provide hydrogencarbonate ions in solution.

13 Assuming that the buffer system is an equilibrium mixture of carbonic acid, water, hydrogencarbonate ions and hydrogen ions, write an equation to represent this.

HINT Reactive Group I metal

What will happen to this system if hydrogen ions are added? Le Châtelier's principle would imply that these hydrogen ions will combine with the hydrogencarbonate ions already present to form carbonic acid.

14 Write a balanced formula equation for the reaction described above.

15 In which direction does this drive the overall equilibrium reaction?

HINT Ensure you know which are reactants and which are products in this reaction.

What will happen to this system if hydroxide ions are added? Le Châtelier's principle would imply that these hydroxide ions will combine with the hydrogen ions already present to form water.

16 Write a balanced formula equation for the reaction described above.

17 In which direction does this drive the overall equilibrium reaction?

Carbonic acid and hydrogen carbonate ions play a part in keeping the pH of sea water constant. Hydrogencarbonate ions are important in keeping the pH of human blood plasma between pH 7.35 and 7.45.

18 What might happen if the pH of human blood plasma falls outside this range?

Alkaline buffer solutions can be formed from a weak base and one of its salts. Ammonia solution ($NH_4OH_{(aq)}$) and ammonium chloride (NH_4Cl) are used.

19 Assuming that this buffer system is an equilibrium mixture of ammonium hydroxide, water, ammonium ions and hydroxide ions, write an equation to represent this.

What will happen to this system if hydrogen ions are added? Le Châtelier's principle would imply that these hydrogen ions will combine with the hydroxide ions already present to form water.

20 Write a balanced formula equation for the reaction described above.

TRAP Don't include the chloride ions – these are 'spectator' ions and don't play a major part in the reaction.

21 In which direction does this drive the overall equilibrium reaction?

What will happen to this system if hydroxide ions are added? Le Châtelier's principle would imply that these hydroxide ions will combine with the ammonium ions already present to form ammonium hydroxide.

22 Write a balanced formula equation for this reaction.

23 In which direction does this drive the overall equilibrium reaction?

24 Give three uses for buffer solutions.

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18 PIPETTES AND BURETTES

PIPETTES

A bulb pipette (Figure 1) is designed to deliver a fixed volume of liquid accurately. It is washed with distilled water, allowed to drain and rinsed before use with the delivery to be pipetted. The tip (bottom) of the pipette is placed into the liquid to be measured out and suction is applied to the top opening of the pipette.

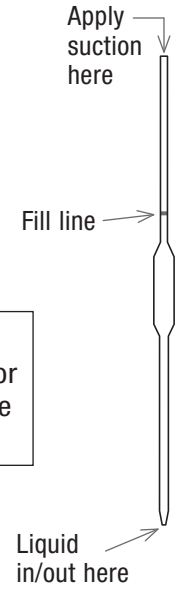

Figure 1

1 Using your mouth to suck liquid into a pipette is not allowed. Give a reason for this.

The liquid is drawn up the pipette. Usually the liquid is filled to just above the fill line (or mark), removed from the liquid and the contents of the pipette are allowed to drain out into another container until the bottom of the meniscus of the liquid is level with the line shown in figure 1.

TRAP It is important to avoid parallax error when checking the level of the liquid.

2 Draw a labelled diagram showing the relationship between the meniscus and the fill line.

3 Explain what is meant by *parallax error*.

A common way to introduce a liquid into a bulb pipette is to use a bulb filler. A bulb filler is made out of rubber or a flexible polymer. It should be resistant to chemicals. Figure 2 shows a typical bulb filler. Details vary but most have three valves (e.g. A, B and C).
The basic sequence is:

- air release (A)
- filling of liquid (B)
- delivery of liquid (C).

The top of the pipette is carefully inserted into the bottom of the filler before use.

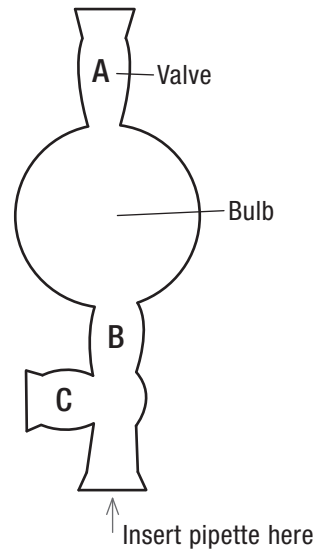

Figure 2

4 Why should you be careful when inserting the pipette into the filler?

5 Study the list of operations involved in filling and discharging (emptying) a bulb pipette. Write them in the correct sequence.

a Remove tip of pipette from liquid.

b Place tip of pipette under surface of liquid.

c Press valve B to draw liquid into pipette.

d Release valve B when liquid is just above fill mark.

e Press valve C until bottom of meniscus reached fill line.

f Squeeze bulb with valve A depressed to remove air from bulb.

g Place pipette over mouth of reaction vessel.

h Place pipette over beaker for waste/excess liquid.

i Place tip of pipette on inside of mouth of reaction vessel to deliver last part of liquid.

j Press valve C until all liquid leaves the pipette.

THINGS TO KNOW Make sure you understand how chemical equipment works and how to use it correctly and accurately.

6 Why should liquid being drawn into pipette not be allowed to go into the filler?

7 After use, why should the bulb in the filler not be allowed to remain 'squeezed in'?

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BURETTES

The burette (figure 3) is designed to deliver variable amounts of liquid accurately. It is basically a hollow tube with a scale on the side, with a tap (stopcock) at the bottom to regulate the flow of the liquid. There are many types of tap – ensure you are familiar with the operation of the one you are using. The most common burettes tend to be 25 mL and 50 mL types. They are usually marked in 0.1 mL divisions and can be read to 0.05 mL by reading between the markings. Before use, they should be rinsed with distilled water, allowed to drain and then rinsed with the solution they are to deliver.

8 Why should a burette first be rinsed with distilled water?

9 Why should the burette then be rinsed with the liquid to be used?

HINT How would any leftover distilled water affect the new solution?

It is best to use a small funnel to fill a burette. There should be a gap between the funnel and the inside walls of the burette to allow air to escape, avoiding overflow. The burette should be filled just past the zero mark. Then put a container under the burette and carefully drain until the zero mark is reached.

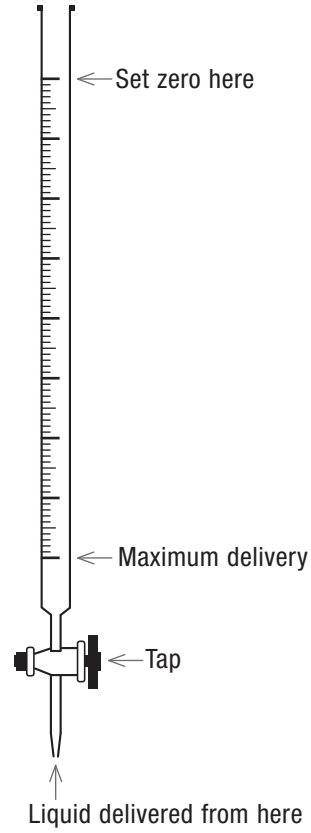

Figure 3

10 Describe how the reading on a burette is taken.

11 Describe how a burette should be set up on a bench to ensure safe and accurate use.

After the burette has been used, wash it with distilled water at least twice. Then store upside down with the tap open.

Figure 4 shows a conical flask. It is used in acid–base titrations involving indicators.

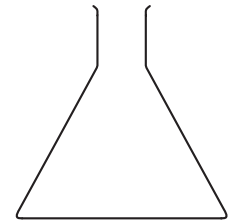

Figure 4

12 Give a definition for *indicator*.

13 Give two examples of indicators.

The conical flask is washed out with distilled water. A pipette is used to deliver a specific volume (usually 25 mL) of one of the reacting solutions into the conical flask. A few drops of indicator are then added to the solution and the flask is swirled to ensure that the indicator is mixed in. A note is made of the colour of the mixture. The mouth of the conical flask is placed under the burette, leaving a gap of two to five centimetres. The first step is a 'rough' titration.

One way to carry out a rough titration (assuming you are right-handed) is listed below.

- Using your left hand, hold tap from opposite side.
- Hold mouth of conical flask with right hand.
- Add solution from burette 1 mL at a time, swirling mixture after each addition.
- When mixture changes colour permanently, cease adding solution from burette.
- Note the quantity of liquid added from the burette.

Another way to carry out a rough titration is to add the solution from the burette slowly, continuously swirling the mixture. When colour change starts to persist, slow the rate of addition from the burette. When the colour change is permanent, cease addition and note the volume added.

14 What has been reached when the indicator changes colour?

15 The value of the rough titration is almost certainly to be greater than the quantity needed for exact neutralisation. Outline the steps you would follow to carry out an accurate titration.

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19 PRIMARY STANDARDS

Quantitative analysis uses solutions of known concentrations to calculate the concentrations of other solutions. An essential part of a quantitative analysis is a primary standard. This is a substance that is used to make a solution of very accurately known concentration. This in turn is used to make other standard solutions.

TRAP Don't confuse qualitative analysis with quantitative analysis. Qualitative analysis tells you the identity of a substance; quantitative analysis tells you how much there is present.

In order for a substance to be suitable for use as a primary standard, it must:

- be available in a pure form
- be very soluble in water
- react with desired substances
- be stable – not react with substances present in air, i.e. oxygen, carbon dioxide and water vapour
- not change mass while being weighed, eg by losing water to the atmosphere (efflorescent) or gaining water from the atmosphere (hygroscopic).

In addition, it is desirable that the substance has a high molar mass so that errors in weighing make as small a percentage difference as possible.

1 Suggest one other advantageous property of a substance used to make a primary standard.

Common primary standards include:

- Sodium carbonate, Na_2CO_3
- Potassium dichromate, $K_2Cr_2O_7$
- Hydrated copper sulfate, $CuSO_4.5H_2O$

2 Calculate the molar masses of the following compounds, given these atomic masses:
Na = 22.9, C = 12.0, O = 16.0, K = 39.1, Cr = 52.0, Cu = 63.5, S = 32.1, H = 1.00.

a Sodium carbonate

b Potassium dichromate

c Hydrated copper sulfate

TRAP Be careful when calculating molar masses. Always check your calculations.

When making standard solutions, the use of pure water (deionised/demineralised or distilled) and clean glassware is essential. Volumetric glassware is designed to measure the volumes of liquids accurately.

3 Why shouldn't you use tap water to make up standard solutions?

4 Name two items of volumetric glassware.

MAKING A PRIMARY STANDARD

Sodium carbonate is often used as a primary standard.

5 Give the formula for sodium carbonate.

It is best to break down calculations into stages. The first step is to calculate the number of moles of sodium carbonate needed to make a solution of the desired concentration.

In figure 1 there are two equations which show the relationship between moles, concentration and volume.

$$c = \frac{n}{v} \qquad n = c \times v$$

c = concentration in moles per litre
n = number of moles
v = volume in litres

Figure 1

After calculating the number of moles, you need to find out what mass this is. Figure 2 shows the relationship between masses and moles.

$$\text{number of moles} = \frac{\text{mass present}}{\text{molar mass}}$$

or

$$\text{mass present} = \text{number of moles} \times \text{molar mass}$$

Figure 2

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WORKED EXAMPLE

What mass of sodium carbonate would be required to make 250 mL of 0.1M solution?

- Formula of sodium carbonate = Na_2CO_3
- Molar mass = 105.8 g

Number of moles required:

- Substitute values in equation $n = cv$

$n = 0.1 \times 0.25 = 0.025$ moles.

Mass required:

- Substitute values in equation

mass present = number of moles × molar mass

$= 0.025 \times 105.8 = 2.645$ g

TRAP Don't forget to change millilitres to litres if necessary.

6 What mass of sodium carbonate would be required to make 500 mL of 0.5M solution? Show your working.

7 What mass of hydrated copper sulfate, $CuSO_4.5H_2O$, would be required to make 250 mL of 0.2M solution? Show you working.

8 Would you use a beaker to measure out the volumes mentioned in questions 6 and 7? Explain your answer.

After calculating the quantities involved, you then need to carry out the preparation of the solution. An essential piece of apparatus is a volumetric flask. (See figure 3.) This is designed to contain an accurate (fixed) volume of liquid at a specific temperature. There is a securely fitting stopper (glass or plastic) which allows the flask to be inverted.

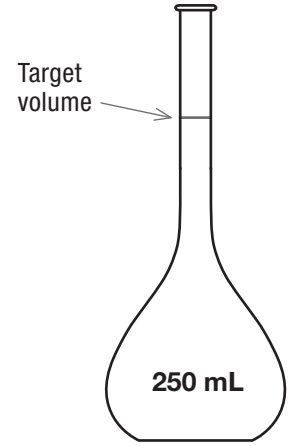

Figure 3

9 Give another reason for a volumetric flask to have a stopper.

A ring is etched into the neck of the volumetric flask. This mark indicates when the correct volume has been reached. In order to ensure a correct reading, the bottom of meniscus of the solution must be level with the mark. To avoid 'parallax error', line your eye up as shown in Figure 4 when taking a reading.

10 A student had to warm a solution in order for all of the solid to dissolve. Why should this solution be allowed to cool to a temperature such as 20°C before being poured into a volumetric flask?

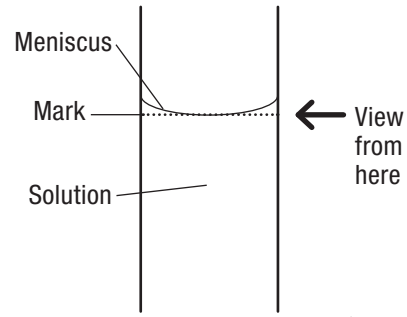

Figure 4

One sequence for making a primary standard is shown below.

- Weigh out a mass as close as possible to that previously calculated.
- Place the solid in the beaker.
- Dissolve in distilled water, using slightly less than the volume required.
- Transfer the solution to a volumetric flask, using a filter funnel to assist.
- Rinse the beaker and filter funnel with distilled water, ensuring that this goes into the flask.
- Top up to the mark with distilled water.
- Stopper flask and invert.
- Calculate molarity using the mass weighed out.

THINGS TO KNOW Make sure you understand how chemical equipment works and how to use it correctly and accurately.

11 Explain why the flask is inverted in the sequence above.

12 A student made 250 mL of a solution containing 1.32 g of sodium carbonate. What is the molarity of this solution? Show your working.

HINT $c = n/v$

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20 ACID AND BASE SOLUTIONS

Acids and bases are used in almost all areas of life. It is important for chemists to know the identity and concentration of these chemicals when we use them. The most common acids used in school laboratories are hydrochloric, nitric and sulfuric acid.

A base is the opposite of an acid. Common bases are sodium hydroxide, potassium hydroxide and ammonium hydroxide.

1 Give the formula for:

a hydrochloric acid. **b** nitric acid.

c sulfuric acid. **d** sodium hydroxide.

e potassium hydroxide. **f** ammonium hydroxide.

2 Outline the similarities and differences between a *base* and an *alkali*.

TRAP Often the term alkali is used interchangeably with base but they are not the same.

The concentration of a solution has to be determined before it can be used in quantitative analysis. Sodium carbonate is used to make solutions (primary standards) against which other solutions are standardised.

3 Outline the difference between *qualitative* and *quantitative* analysis.

4 Name the salts produced by:

a hydrochloric acid. **b** nitric acid. **c** sulfuric acid.

One possible sequence in quantitative analysis is shown below.

- Work out an equation for the reaction.
- Study the ratios of reactants.
- Carry out a titration.
- Calculate concentrations.

WORKED EXAMPLE

Write a balanced formula equation for the reaction between hydrochloric acid and sodium carbonate.

- The first step is to write a word equation:

  hydrochloric acid + sodium carbonate → sodium chloride + carbon dioxide + water

- Next, substitute the formulae in the word equation.

  $HCl_{(aq)} + Na_2CO_{3(aq)} \rightarrow NaCl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$

- Then balance the formula equation.

  $2HCl_{(aq)} + Na_2CO_{3(aq)} \rightarrow 2NaCl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$

5 What is the ratio of hydrochloric acid to sodium carbonate in the reaction shown in the worked example?

6 Write a word equation and then a balanced formula equation for the reaction between nitric acid and sodium carbonate.

HINT Acid + carbonate → salt + carbon dioxide + water.

7 Write a word equation and then a balanced formula equation for the reaction between sulfuric acid and sodium carbonate.

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21 TITRATION

Titration involves expertise in many areas of chemistry.
Common errors occur in:

- balancing equations
- calculating concentrations
- manipulating apparatus.

TRAP If you do not balance equations correctly, this will affect your final results.
Be especially careful with substances that generate more than one mole of hydrogen ions or hydroxide ions per mole when dissolved.

1 Give the formulae of the following.

a phosphoric acid

b calcium hydroxide

2 Write a word equation and then a balanced formula equation for the reaction between phosphoric acid and sodium hydroxide solution.

HINT Acid + base → salt + water

3 Write a word equation and then a balanced formula equation for the reaction between nitric acid and calcium hydroxide solution.

4 Write a word equation and then a balanced formula equation for the reaction between sulfuric acid and ammonium hydroxide.

Molarity is the concentration of a solution expressed in moles per litre (M or mol L^{-1}).

In any molarity calculation, you will be given two of the quantities and asked to calculate the third. You will need to perform some basic algebra to decide which formula to use.

5 Study Figure 1. Copy and complete parts (b) and (c).

(a) $\text{concentration} = \frac{\text{moles}}{\text{litres}}$

(b) litres =

(c) moles =

Figure 1

WORKED EXAMPLE

What volume of liquid would be required to produce a solution of 0.5 mol L^{-1} given 0.25 moles of solute?

- Use the formula: litres (volume) = moles ÷ concentration.
- Substitute values into the formula: litres = 0.25 ÷ 0.5
- Perform the calculation. litres = 0.125
- Check units: 0.125 L

6 What volume of liquid would be required to produce a solution of 0.2 mol L^{-1} given 0.5 moles of solute? Show your working.

HINT Ensure that your answer stands out so that the marker can see it easily; e.g. underline: <u>0.125 L</u>.

7 How many moles of solute would be required to produce a solution of 0.1 mol L^{-1} given 0.2 litres of solution? Show your working.

8 What would be the concentration of a solution given 0.25 litres of solution and 0.5 moles of solute? Show your working.

HINT If you have problems in remembering how to work out concentrations, consider:
Concentration = moles per litre
Replace the 'per' with 'divide' and you get:
concentration = moles ÷ litres

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Titration is a technique used to find the concentration of one solution by reacting it with an accurately determined volume of a solution of known concentration. The equivalence point is when the exact quantities of reactants are involved to conform to the equation for the reaction. When a reaction is complete, there is often a physical change (end point) that can be detected. Titrations involving acid–base neutralisation are very common. The most common changes studied are:

- pH
- temperature
- conductivity.

In all cases a fixed volume of one solution (eg base) is added to a container and the other solution (eg acid) added in increments (small portions) and the change of pH is measured and compared to the volume added.

TRAP

Make sure you have a sound understanding of the concept of **equivalence point**. The equivalence point is where the reaction has reached a specific stoichiometric ratio of reactants. In acids and bases the equivalence point is reached when the number of H^+ ions equals the number of OH^- ions.

9 Name the piece of glassware most likely to be used to measure out a fixed volume of reactant solution.

10 Name the piece of glassware most likely to be used to measure out variable volumes of reactant solution.

THINGS TO KNOW

There are many Chemistry questions that involve an understanding of how pH changes when acids and bases are added to neutralise each other in a titration.

Ensure you know:

- the names and formulae of reactants
- how to write a balanced equation for the reaction
- whether the acids and bases involved are weak or strong
- whether the acid is added to the base or vice versa.

11 Copy and complete the sentence:

Strong acids and bases dissociate _____ in water, whereas _____ acids and bases only partially dissociate.

TRAP

pH is a mathematical expression describing the hydrogen ion concentration of a solution. At the equivalence point in an acid–base titration, the pH changes rapidly. One common misconception is that the equivalence point always occurs at pH 7.0. This only occurs with strong acid/strong base and weak acid/weak base pairs. The equivalence point of a titration involving a strong acid/weak base will be under pH 7.0.

12 What will be the approximate pH of the equivalence point of a titration involving a weak acid/strong base?

If using an indicator, it is important to use one that changes colour (end point) when the equivalence point is reached. pH changes can be monitored using a pH meter, pH probe with a data logger or an indicator. Any electronic device used to measure pH needs to be calibrated. The probe is placed into a solution of known pH (a buffer solution) and the device adjusted to display this pH. After rinsing in distilled water, the probe is then placed into one of the solutions.

13 Why would the probe be rinsed with distilled water after calibration?

Things to know The initial pH is noted and continues to be recorded as the other solution is added from a burette. The equivalence point is indicated by a rapid change in pH. Combinations of strong acids and strong bases give steeper changes in pH than combinations of weak acids and weak bases.

14 Give a definition for pH.

15 Compare the equivalence point with the end point in a titration.

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22 TITRATION CURVES

If you plot the pH of a solution against the volume of acid or base added to it, you will get a pH (titration) curve. Figure 1 shows the titration curve obtained when a strong base is added to a strong acid. Note the rapid change in pH at the equivalence point.

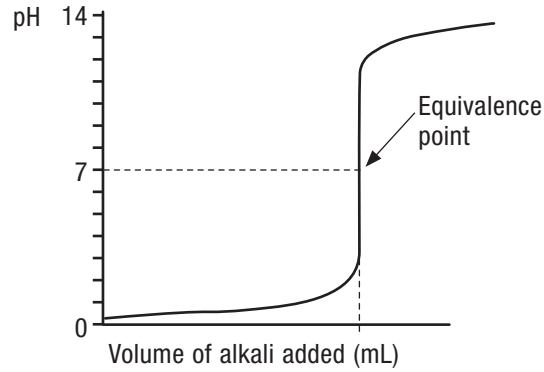

Figure 1

1 Name a typical strong acid.

2 Name a typical strong base.

3 Draw a diagram similar to figure 1 to show what happens when a strong acid is added to a strong base.

THINGS TO KNOW

Different indicators change colour over different pH ranges – and not necessarily at pH 7.

Figure 2 shows the titration curve obtained when a strong acid is added to a weak base. Note, in comparison to figure 1, the different initial shape of the curve and the less rapid change in pH at the equivalence point.

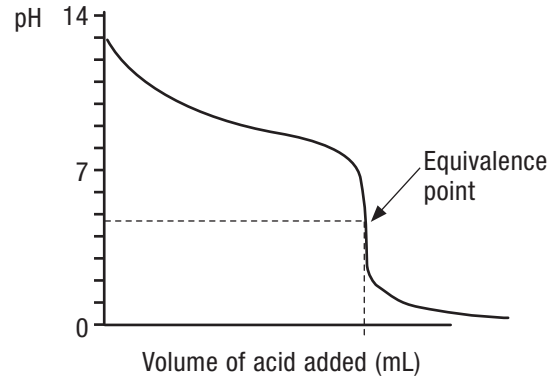

Figure 2

4 Name a typical weak base.

5 What would be the pH at the equivalence point for the reaction shown in figure 2?

6 Draw a diagram similar to figure 2 to show what happens when a strong base is added to a weak acid.

Study figure 3.

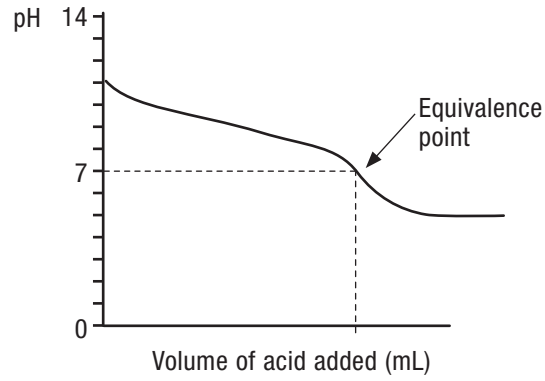

Figure 3

7 What is the pH of the equivalence point?

8 Identify the two types of chemicals being mixed here.

9 Comment on the shape of the curve.

We need to make sure that the end point is as close to the equivalence point as possible. This means we must use an indicator that changes colour at or very near the pH of the equivalence point.

TRAP

Don't confuse **equivalence point** and **end point**.

Equivalence point: The solutions have been mixed in exactly the right quantities according to the equation.

End point: When the indicator changes colour – not necessarily exactly the same as the equivalence point.

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23 TITRATION CALCULATIONS

THINGS TO KNOW

Before you start this unit, make sure you are familiar with the formulae of common acids and bases (alkalis) and calculations involving molarity.

In titrations, there are four quantities involved:

- volume of acidic solution
- concentration of acidic solution
- volume of basic solution
- concentration of basic solution.

You will be given values for three of these quantities and be required to calculate the fourth quantity (usually the concentration).

1 In calculations involving molarity, you will have encountered the equation $n = c \times v$. Explain in words what this equation means.

SOLVING TITRATION CALCULATIONS:

TRICK

Use the following sequence to help you solve titration calculation questions.

After a few questions, you will find that you can streamline this process.

- Identify the reactants and quantities involved.
- Identify the item to be determined.
- Write a word equation showing reactants and products.
- Write a balanced formula equation.
- Check ratio of acid : base.
- Do calculations. Don't forget to change volumes to litres.

WORKED EXAMPLE

25 mL of sodium hydroxide was titrated with a 0.1 mol L^{-1} solution of hydrochloric acid. 15.3 mL of the acid was required for neutralisation. Calculate the molarity of the sodium hydroxide solution.

- Sodium hydroxide (NaOH) 25 mL and hydrochloric acid (HCl) 15.3 mL and 0.1 mol L^{-1}
- The concentration of the sodium hydroxide solution needs to be calculated.
- Sodium hydroxide + hydrochloric acid → sodium chloride + water
- $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O$
- Ratio is 1:1, so number of moles of sodium hydroxide and hydrochloric acid will be the same.
- Because number of moles (n) is the same, for acid and base the product of concentration (c) and volume (v) must be the same.

  c(base) × v(base) = c(acid) × v(acid)

  c(base) × 25/1000 L = 0.1 mol L^{-1} × 15.3/1000 L
- Hence, c(base) = (0.1 mol L^{-1} × 15.3/1000 L) ÷ 25/1000 L

  = **0.061 2 mol L^{-1}**

TRICK

It's always a good idea to 'guesstimate' if your answer is appropriate. In this case the sodium hydroxide needed less hydrochloric acid to neutralise it, so the sodium hydroxide must be less concentrated than the hydrochloric acid.

2 25 mL of potassium hydroxide was titrated with a 0.2 mol L^{-1} solution of hydrochloric acid. 31.2 mL of the acid was required for neutralisation. Calculate the molarity of the potassium hydroxide solution.

3 10 mL of nitric acid was titrated with 14.9 mL of a 0.5 mol L^{-1} solution of potassium hydroxide. Calculate the molarity of the nitric acid solution.

4 10 mL of ethanoic acid (CH_3COOH) was titrated with 27.3 mL of a 1.0 mol L^{-1} solution of ammonium hydroxide. Calculate the molarity of the ethanoic acid solution.

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TRAP

Ratios and titration calculations. You need to be careful if you encounter acids and bases that give more than one H^+/OH^- ion per molecule. Some acids, such as hydrochloric (HCl), nitric (HNO_3) and ethanoic (CH_3COOH) acids, are monoprotic (monobasic); i.e. only give one proton (hydrogen ion) per molecule. Sulfuric acid (H_2SO_4) and ethanedioc acid ($(COOH)_2$) are diprotic (dibasic) and give two protons per molecule. Diprotic acids act as if they are twice as concentrated as their nominal concentration.

5 Phosphoric acid has the formula H_3PO_4 and when fully dissociated is described as triprotic. What does *triprotic* mean?

Sodium hydroxide (NaOH), potassium hydroxide (KOH) and ammonium hydroxide (NH_4OH) only give one hydroxide ion per molecule. Calcium hydroxide ($Ca(OH)_2$) gives two hydroxide ions per molecule.

WORKED EXAMPLE

10 mL of sulfuric acid was titrated with 22.8 mL of a 1.0 mol L^{-1} solution of sodium hydroxide. Calculate the molarity of the sulfuric acid solution.

- Sulfuric acid (H_2SO_4) 10 mL and sodium hydroxide (NaOH) 22.8 mL and 1.0 mol L^{-1}
- The concentration of the sulfuric acid solution needs to be calculated.
- Sulfuric acid + sodium hydroxide → sodium sulfate + water
- $H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O$
- Ratio is 1:2, so number of moles of sulfuric acid molecules will be half that of sodium hydroxide. (Number of moles of hydrogen ions and hydroxide ions will be equal.)
- Because number of moles (n) is not the same, we need to adjust the ratio.
- Substitute values, do calculation and then multiply the calculated concentration of sulfuric acid by 0.5 at the end.
- c(acid) × v(acid) = c(base) × v(base)
  (acid) × 10/1000 L = 0.1 mol L^{-1} × 22.8/1000 L
- Hence, c(acid) = (0.1 mol L^{-1} × 22.8/1000 L) ÷ 10/1000 L = 0.228 mol L^{-1}
- Concentration of sulfuric aid = 0.228 × 0.5
  = **0.114 mol L^{-1}**

HINT

Make sure you can follow the logic of the previous calculation before moving on to the questions below. If you find the sequence difficult, try constructing a flow chart to show the steps to take.

6 25 mL of potassium hydroxide was titrated with a 0.2 mol L^{-1} solution of sulfuric acid. 19.1 mL of the acid was required for neutralisation. Calculate the molarity of the potassium hydroxide solution.

7 10 mL of hydrochloric acid was titrated with 43.9 mL of a 0.5 mol L^{-1} solution of calcium hydroxide. Calculate the molarity of the hydrochloric acid solution.

8 10 mL of ammonium hydroxide was titrated with 17.4 mL of a 1.0 mol L^{-1} solution of phosphoric acid. Calculate the molarity of the ammonium hydroxide solution.

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24 CONJUGATE ACID-BASE PAIRS

The **theory of proton transfer** was developed independently by Brönsted and Lowry. It is a very useful theory for accounting for the properties of acids and bases**.** According to Brönsted–Lowry, an acid is a proton donor and a base is a proton acceptor. In aqueous solutions, hydrogen ions are hydrated – i.e. exist as H_3O^+ (hydroxonium) ions.

Using HA to represent a typical (monobasic) acid, we can show how an acid behaves when it is dissolved.

$$HA_{(aq)} + H_2O_{(l)} \rightarrow A^-_{(aq)} + H_3O^+_{(aq)}$$

The HA has acted as an acid because it has lost a proton. The H_2O has acted as a base because it has gained a proton. However, this is an equilibrium process and so is written as:

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons A^-_{(aq)} + H_3O^+_{(aq)}$$

Hence the reaction $H_3O^+_{(aq)} + A^-_{(aq)} \rightarrow H_2O_{(l)} + HA_{(aq)}$ also occurs.

1 In the reaction $H_3O^+_{(aq)} + A^-_{(aq)} \rightarrow H_2O_{(l)} + HA_{(aq)}$

- **a** is $H_3O^+_{(aq)}$ acting as an acid or a base?
- **b** is $A^-_{(aq)}$ acting as an acid or a base?

Figure 1 summarises the acids and bases present in this equilibrium system. Some species on opposite sides of the equation (HA/A^- and H_2O/H_3O^+) only differ by a proton (hydrogen ion). We call these **conjugate acid–base pairs**. The conjugate base of the acid HA is A^- and the conjugate acid of the base H_2O is H_3O^+.

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons A^-_{(aq)} + H_3O^+_{(aq)}$$

Acid Base Base Acid

Figure 1

We can now apply this to the equilibrium system as shown in figure 2.

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons A^-_{(aq)} + H_3O^+_{(aq)}$$

Acid 1 Base 2 Base 1 Acid 2

Figure 2

2 Follow the process outlined above for the actual acid hydrochloric acid, $HCl_{(aq)}$.

- **a** First write a balanced formula equation showing the species that exist when hydrochloric acid dissolves in water.
- **b** Identify the acid–base conjugate pairs.
- **c** Rewrite the equation showing the numbered acid–base pairs using figure 2 as a model.

3 Follow the same process for nitric acid.

- **a** Write a balanced formula equation showing the species that exist when nitric acid dissolves in water.
- **b** Identify the acid–base conjugate pairs.
- **c** Rewrite the equation showing the numbered acid–base pairs using figure 2 as a model.

4 What type of acid is described as *monoprotic* (or *monobasic*)?

5 There have been many definitions of an acid. Find out how the scientists listed below decided if a substance was an acid.

- **a** Antoine Lavoisier (1743–94)
- **b** Humphrey Davy (1778–1829)
- **c** Svante Arrhenius (1859–1927)

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A SELECTED ANSWERS

1 Introduction to organic chemistry

1 Basically correct, if incomplete; many non-living substances contain carbon

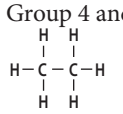
2 Has a valency of four, forms four bonds

3 Covalent; has four outer electrons – more likely to share than transfer electrons

4 Also in Group 4 and can be tetravalent

5 CH_4

6 C_2H_6



7 Missing formulae: C_3H_8, C_5H_{12}, C_7H_{16}, C_8H_{18}, $C_{10}H_{22}$

8 Pentane to decane

9 As the number of atoms increases, so does the melting point and boiling point

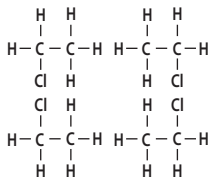
10 They all end in '-ane' **11** $C_{11}H_{24}$ and $C_{12}H_{26}$

12 Answer should follow trend, i.e. increase. Actual figures are -26°C, 196°C and -9°C, 216°C

2 Nomenclature

2 Group VII elements **3** CH_3Cl **4** CH_2Cl_2

5



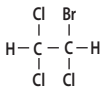
6 $CHCl_3$ **7** *mono* = one

9 How long carbon chain is, any branching **10** $C_2H_4Cl_2$

11 **a** 1, 1-dibromo 2-chloroethane, $C_2H_3Br_2Cl$.

b 1, 1-dichloro 2-bromoethane, $C_2H_3BrCl_2$

c



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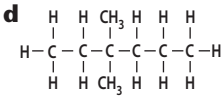
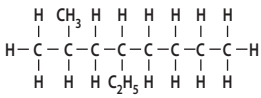


12 **a** C_6H_{14} **b** 2-methylpentane

13 **a** 2,3-dimethylbuttane **b** 2, 2-dimethylbutane

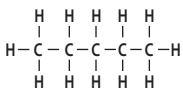
c 3-methylpentane

14



3 Isomerism

1 e.g. hexane

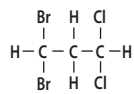


2 **a** 1, 1-dichloro 2, 3-dibromopropane

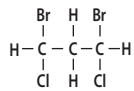
b 1, 1-dichloro 2, 2-dibromopropane

c 1, 1-dichloro 1, 3-dibromopropane

3 e.g. 1, 1-dibromo 3, 3 dichloropropane



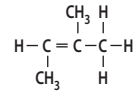
4 e.g. 1, 3-dibromo 1, 3 dichloropropane

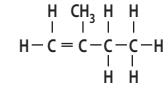


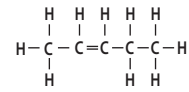
4 Alkenes and alkynes

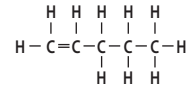
1 Has only one carbon-to-carbon double bond **2** Isomers

3 e.g. pent-1-ene, pent-2-ene, 1, 2 –dimethyl prop-1 ene, 2-methyl but-1-ene.







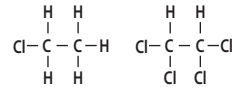


5 Alkynes

1 Ethyne (gas) burns in oxygen (gas) to give carbon dioxide (gas) and water (gas)

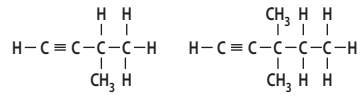
$C_2H_{4(g)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(g)}$

2



3 **a** butyne (but-1-yne) **b** butyne (but-1-yne) **c** but-2-yne

4



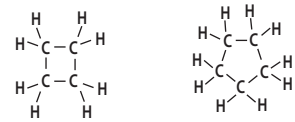
5 They are 'saturated'; i.e. have no carbon-carbon double bonds or carbon-carbon triple bonds.

6 Cyclic hydrocarbons

1 C_3H_6

2 The angle is very different to standard, the bond will be under strain and very reactive.

3



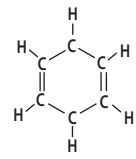
5 The bond angle of cyclopentane (108°) is closer to the 'ideal' (109.5°) than that of cyclobutene (90°) and hence cyclopentane will be more stable.

6 C_6H_{12}

7 C_3H_4 (cyclopropene), C_4H_6 (cyclobutene), C_5H_8 (cyclopentene), C_6H_{10} (cyclohexene)

8 Carbon-carbon double bonds, hence will react with each other.

9



10 Yes, same molecular formulae but different structural formulae.

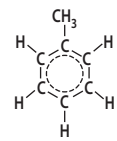
11 Not all carbon-to-carbon bonds are single.

12 Cyclohexa-1, 3, 5 triene **13** Can cause cancer

14 Has a low boiling point, turns to a liquid easily

15 parts per million **16** Lasts a long time

17



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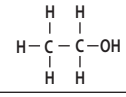
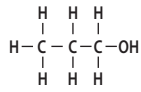
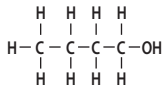
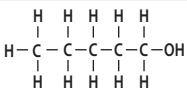
18 C_7H_8

19 He came up with the ring structure showing alternating single and double bonds. This apparently came to him in a dream about a snake holding its own tail.

7 Alkanols

1 CH_4O

2

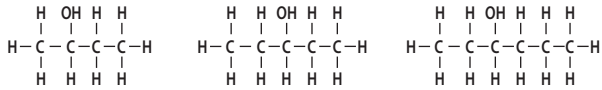
Name	Structural formula	Molecular formula
Ethanol		C_2H_6O
Propanol		C_3H_8O
Butanol		$C_4H_{10}O$
Pentanol		$C_5H_{12}O$

3 Has features of both. Two carbon chain (and hydrogens) of ethane and polar –OH group of water.

4 $C_6H_{12}O_{6(aq)} \rightarrow 2C_2H_5OH_{(aq)} + 2CO_{2(g)}$ **5** Solvent, beverage

6 $C_2H_5OH_{(l)} + 3O_2 \rightarrow 2CO_{2(g)} + 3H_2O$

7 Petrol containing 10% ethanol **8** C_3H_8O **9** C_3H_8O

10 

11 Ethan-1,2 diol **12** Propan-1, 2, 3 triol

13 Increased hydrogen bonding causes an increase in melting and boiling points, also increases viscosity.

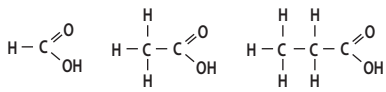
14 Benzene

15 From left to right: hydroxyl group, ring structure, hydrocarbon tail

16 No, the hydroxy group is attached to a ring.

17 Raw material **18** Chemical reactions occurring in the body

8 Organic acids

1 

2 Flavouring, preservative **3** Ethandioic acid **4** $C_2H_2O_4$

5 Flavouring **6** $C_6H_8O_7$

7 Only three hydroxyl groups are attached to C=O (carbonyl) group.

8 $C_7H_6O_2$ **9** Has a larger non-polar part (benzene ring)

10 **a** Ethanoates **b** Propanoates **c** Benzoates

9 Amino acids and proteins

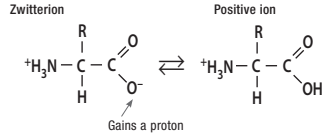
1 **a** $-NH_2$ **b** –COOH **2** Derived from an alkane

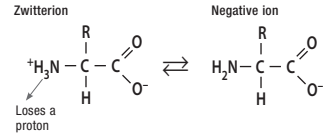
3 $C_2H_5O_2N$

4 Ends of molecules have different electrical charges.

5 Acids donate protons, bases accept protons.

6 Can internally transfer a proton, forming a charged molecule.

7 

8 

9 Zwitterion **10** Dehydration **12** Polypeptide **13** Two

14 Identity of amino acids present and their sequence.

15 Amino acids **16** **a** H^+ **b** Alkyl groups **17** Digestion

18 Identify groups, where they are linked.

19 Much less complicated.

20 **a** The carbonyl groups are relatively negatively charged and the hydrogens are relatively positive. These attract each other. **b** Much less. **c** C=O δ^- δ^+H

21 Chemistry concerned with legal matters.

22 Hydrogen bonds. **23** Structure becomes disorganised.

24 Carries oxygen in the bloodstream.

25 Fibrous: long strips, tough, insoluble, make up connective tissue. Globular: spherical, easily denatured, soluble, specialised function.

26 Enzyme **27** Nitrogen

28 Don't allow to contact skin, wear gloves, use in fume cupboard, make sure no naked flames present, wear goggles, wear lab coat.

29 $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

30 **a** Dangerous **b** Can eat away substances **c** Poisonous **d** Taken in by body **e** Cupboard with extraction of air.

31 Benedict's solution identifies reducing sugars by changing from blue to reddish. Biuret test identifies proteins by changing from blue to violet. Ninhydrin identifies amino acids by changing to a blue/violet colour.

10 Adeniosine triphosphate (ATP)

1 Figure 2b has 'lost' four hydrogen atoms and has four negative charges.

2 Phosphate **3** Exothermic **4** Adenosine diphosphate

5 Organic catalyst, lowers activation energy

6 $ADP + PO_4^{3-} \rightarrow ATP + H_2O$

7 $C_6H_{12}O_{6(aq)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)}$ $\Delta H = -2\,900$ kJ

8 In the absence of oxygen **9** Potential **10** Cells

11 Sum of all of the chemical reactions occurring in the body

11 Calorimetry

1 Container in which experiments measuring heat changes are carried out.

2 Reactants are at a lower energy level than products. Heat is taken in from surroundings. ΔH is positive. Temperature of the system falls and then rises.

3 **a** Enthalpy change, joules **b** Specific heat, $°C.g^{-1}$ **c** Change in temperature, °C or K

4 $\Delta H = mC\Delta T$.
$\Delta H = 150\ g \times 4.2\ Jg^{-1} \times 10.2°C = -6426\ J = -6.43\ kJ$
No. of moles = $75/1000 \times 2.0 = 0.15$, hence $\Delta H = -42.9\ kJ\ mol^{-1}$

5 Heat is generated inside the calorimeter and polystyrene is a better insulator

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12 Enthalpy of combustion

1 Styrofoam would melt/burn. Metal is a better conductor.

2 No, some is lost to surroundings.

3 $\Delta H = mC\Delta T$. Need to calculate heat absorbed by calorimeter plus water.
$(37.4 \times 0.39 \times 21.7) + (100 \times 4.2 \times 21.7) = 317\text{ J} + 9114\text{ J} = 9431\text{ J}$. This is for 1.1 g, hence for 1.0 g = 8574 J

4 2.5 g of propanol was burned.
Change in temperature was 40.8°C.
Calorimeter ($34.1 \times 0.9 \times 40.8 = 1252$ J).
Water ($400 \times 4.2 \times 40.8 = 68\,544$ J. Total 69 796 J = 69.8 kJ.
Mass of one mole of propanol (C_3H_8O) = 64.1.
Hence 2.5 g = 0.04 mole. Δ= -1745kJ mol^{-1}.
(Actual value is around 2000 kJ mol^{-1})

13 Enzymes

1 Common, not according to IUPAC

3 The substrate 'fits' into the binding site due to its shape and structure

4 Structure become disorganised

5 Its shape and structure have been changed.

6 Pepsin's maximum activity occurs around pH 2, salivary amylase at around pH 7 and amylase at around pH 10.

7 Both speed up reactions, inorganic catalysts are not as specific as enzymes and work over a wider range of temperature and pH

14 Equilibrium reactions

1 $\rightleftharpoons$

2 Teacher to check. Discussion should mention that reversible reactions can only go one way at a time, but direction can be changed with changed conditions. In an equilibrium reaction, forward and reverse reactions occur simultaneously.

3 Only a small amount of reactants have changed to products.

4 The mixture of reactants and products.

5 Whether you start with 100% reactants or 100% products, eventually the system will come to the same equilibrium point, i.e. same ratio of reactants:products.

6 **a** $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
b $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ **c** $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$
d $CH_3OH_{(l)} + CH_3COOH_{(l)} \rightleftharpoons CH_3COOCH_{3(l)} + H_2O_{(l)}$

7 No. Catalysts only allow reactions to reach equilibrium more quickly.

9 The table tennis model is not accurate. The escalator model more closely matches what happens.

15 Further equilibrium reactions

1 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

2 **a** Drive position of equilibrium to right, increase ratio of products:reactants. **b** Drive position of equilibrium to right, increase ratio of products:reactants.

3 Nitrogen and hydrogen react to form ammonia.

4 **a** Drive position of equilibrium to left, decrease ratio of products:reactants. **b** Drive position of equilibrium to left, decrease ratio of products:reactants.

5 $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CH_3O_{(g)}$

6 **a** Drive position of equilibrium to right, increase ratio of products:reactants. **b** Drive position of equilibrium to right, increase ratio of products:reactants.

7 Gases, changes, equilibrium, pressure.

8 Only reactions where reactants and products have a different volume

9 **a** Yes. Volume of reactants negligible (solid), much less than volume of products. **b** No. Ratio is 1:1, no difference in volume. **c** Yes, ratio is 1:2.

16 Calculating pH

1 As hydrogen ions concentration increases, the pH decreases. pH is the negative value of the power of ten of the concentration in $mol.L^{-1}$

2 **a** 1.7 **b** 0.82 **c** 1.7 (Remember that 0.1 $mol.L^{-1}$ sulfuric acid will give 0.2 $mol.L^{-1}$ equivalent.)

3 **a** 7.9×10^{-5} **b** 10^{-13} **c** 3.55×10^{-8} **d** 7.9×10^{-2}

17 Buffer solutions

1 Ethanoic, citric, phosphoric

2 Ammonium hydroxide, calcium hydroxide

3 An equilibrium system will tend to react to minimise any changes imposed on the system.

4 The left. Weak acid and only partially dissociates.

5 It would dissociate almost completely and equilibrium would lie to the right.

6 One that doesn't take part in the reaction, it is there at the beginning and the end, unchanged.

7 $\underset{\text{Acid 1}}{CH_3COOH_{(l)}} + \underset{\text{Base 2}}{H_2O_{(l)}} \rightleftharpoons \underset{\text{Base 1}}{CH_3COO^-{}_{(aq)}} + \underset{\text{Acid 2}}{H_3O^+{}_{(aq)}}$

8 $CH_3COO^-{}_{(aq)} + H_3O^+{}_{(aq)} \rightleftharpoons CH_3COOH_{(l)} + H_2O_{(l)}$

9 To the left **10** $H_3O^+{}_{(aq)} + OH^-{}_{(aq)} \rightleftharpoons 2H_2O_{(l)}$

11 To the right **12** Sodium hydrogencarbonate

13 $H_2CO_{3(aq)} + H_2O_{(l)} \rightleftharpoons H_3O^+{}_{(aq)} + HCO_3^-{}_{(aq)}$

14 $H_3O^+{}_{(aq)} + HCO_3^-{}_{(aq)} \rightleftharpoons H_2CO_{3(aq)} + H_2O_{(l)}$ **15** To the left

16 $H_3O^+{}_{(aq)} + OH^-{}_{(aq)} \rightleftharpoons 2H_2O_{(l)}$ **17** To the right

18 Normal cell reactions don't occur. Metabolism will be affected, death could result

19 $NH_4OH_{(aq)} + H_2O_{(l)} \rightleftharpoons NH_4^+{}_{(aq)} + OH^-{}_{(aq)}$

20 $H_3O^+{}_{(aq)} + OH^-{}_{(aq)} \rightleftharpoons 2H_2O_{(l)}$ **21** To the right

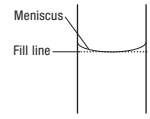
22 $NH_4^+{}_{(aq)} + OH^-{}_{(aq)} \rightleftharpoons NH_4OH_{(aq)}$ **23** To the left

24 Calibrate pH meters, maintain optimum pH for industrial processes, allow enzymes to work in biological/metabolic reactions

18 Pipettes and burettes

1 Liquid could be toxic/corrosive.

2



3 Looking at line at an angle which results in an inaccurate reading.

4 Pipette is made of glass and can break.

5 f, b, c, d, a, e, g, j, i, h.

6 Could damage filler (especially valves).

7 Bulb can stay depressed. **8** Ensure no impurities left.

9 So that liquid is not diluted.

10 Look at the meniscus. Numbers increase as you move down the burette.

11 Use retort stand and clamp to ensure burette is level.

12 Substance that changes colour when there is a change, such as in pH

13 Litmus, phenolphthalein, methyl orange **14** End-point

15 Repeat, but stop a few mL short of previous end point. Then add solution from burette a few drops at a time (swirling after

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each addition) until get colour change. Repeat process until get two titrations that agree to within 0.05 mL of each other.

19 Primary standards

1 Have a known formula/not toxic.

2 **a** 105.8 g **b** 294.2 g **c** 249.6 g **3** It isn't pure.

4 Pipette, burette, measuring cylinder, volumetric flask

5 Na_2CO_3

6 $n = 0.5 \times 0.5 = 0.25$ moles. Mass $= 0.25 \times 105.8 = 26.45$ g

7 $0.25 \times 0.2 \times 249.6 = 12.49$ g

8 No. Beaker only gives approximate volumes. Would need to use e.g. volumetric flask.

9 Stop anything entering, stop water evaporating over time.

10 The nominal volume is set for a particular temperature (e.g. 20°C). Will give a wrong reading if not at this temperature.

11 Ensure contents are mixed.

12 $1.32/105.8 = 0.0125$ moles. $c = 0.0125/0.25 = 0.0499$ mol.L^{-1}

20 Acid and base solution

1 **a** HCl (aq) **b** HNO_3 **c** H_2SO_4 **d** NaOH **e** KOH **f** NH_4OH

2 Both are the opposites of acids, alkalis are soluble bases.

3 Qualitative – what is present. Quantitative – how much is present

4 **a** Chlorides **b** Nitrates **c** Sulfates **5** 2:1

6 Nitric acid + sodium carbonate → sodium nitrate + carbon dioxide + water
$2HNO_{3(aq)} + Na_2CO_{3(s)} \rightarrow 2NaNO_{3(aq)} + CO_{2(g)} + 2H_2O_{(l)}$

7 Sulfuric acid + sodium carbonate → sodium sulfate + carbon dioxide + water
$H_2SO_{4(aq)} + Na_2CO_{3(s)} \rightarrow Na_2SO_{4(aq)} + CO_{2(g)} + H_2O_{(l)}$

21 Titration

1 **a** H_3PO_4 **b** $Ca(OH)_2$

2 Phosphoric acid + sodium hydroxide → sodium phosphate + water
$H_3PO_{4(aq)} + 3NaOH_{(aq)} \rightarrow Na_3PO_{4(aq)} + 3H_2O_{(l)}$

3 Nitric acid + calcium hydroxide → calcium nitrate + water
$2HNO_{3(aq)} + Ca(OH)_{2(aq)} \rightarrow Ca(NO_3)_{2(aq)} + 2H_2O_{(l)}$

4 Sulfuric acid + ammonium hydroxide → ammonium sulfate + water
$H_2SO_{4(aq)} + 2NH_4OH_{(aq)} \rightarrow (NH_4)_2SO_{4(aq)} + 2H_2O_{(l)}$

5 **b** moles/concn. **c** concn. × litres

6 $0.5/0.2 = 2.5$ L **7** $0.1 \times 0.2 = 0.02$ moles

8 $0.5/0.25 = 2$ mol.L^{-1} **9** Pipette **10** Burette

11 Completely, weak **12** Less than pH7

13 So none of the buffer solution is carried into the solution under test.

14 Negative log (base 10) of the hydrogen ion concentration (moles per litre).

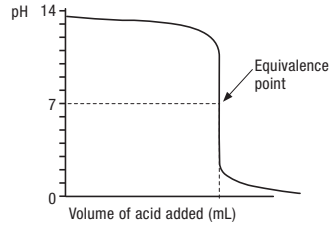
15 Equivalence point: when reactants are exactly at ratio specified by stoichiometry. End point: when indicator changes colour.

22 Titration curves

1 Sulfuric, nitric, hydrochloric

2 Sodium hydroxide, potassium hydroxide

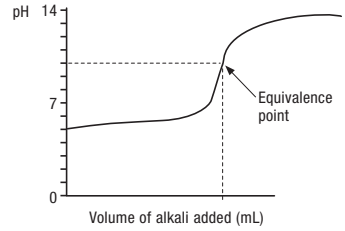
3



4 Calcium hydroxide, ammonium hydroxide

5 Approximately pH4 to pH5

6



7 pH7 **8** A weak acid is being added to a weak base.

9 Gradual, much less severe than others.

23 Titration calculations

1 The number of moles equals the concentration (moles per litre) multiplied by the volume (litres).

2 $25/1000 \times c = 31.2/1000 \times 0.2$.
$c = 31.2/25.0 \times 0.2 = 0.25$ mol.L^{-1}

3 $c = 14.9/10.0 \times 0.5 = 0.745$ mol.L^{-1}

4 $c = 27.3/10 \times 1.0 = 2.73$ mol.L^{-1}

5 Gives three hydrogen ions (protons)

6 $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$
$c = 19.1/25 \times 0.2 = 0.1528$ Need to adjust for stoichiometry $(0.1528 \times 0.5) = 0.0764$ mol.L^{-1}

7 $2HCl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O$
$c = 43.9/10 \times 0.5\ (\times 2.0) = 4.39$ mol.L^{-1}

8 $3NH_4OH + H_3PO_4 \rightarrow (NH_4)_3PO_4 + 3H_2O$.
$c = 17.4/10 \times 0.1\ (\times 3) = 5.22$ mol.L^{-1}

24 Conjugate acid–base pairs

1 **a** An acid **b** A base

2 **a** $HCl_{(aq)} + H_2O_{(l)} \rightleftharpoons Cl^-_{(aq)} + H_3O^+_{(aq)}$
b HCl/Cl^-, H_3O^+/H_2O
c Acid 1 + Base 2 ⇌ Base 1 + Acid 2
$HCl_{(aq)} + H_2O_{(l)} \rightleftharpoons Cl^-_{(aq)} + H_3O^+_{(aq)}$

3 **a** $HNO_{3(aq)} + H_2O_{(l)} \rightleftharpoons NO_3^-{}_{(aq)} + H_3O^+_{(aq)}$
b HNO_3/NO_3^-, H_3O^+/H_2O
c Acid 1 + Base 2 ⇌ Base 1 + Acid 2
$HNO_{3(aq)} + H_2O_{(l)} \rightleftharpoons NO_3^-{}_{(aq)} + H_3O^+_{(aq)}$

4 Only give one hydrogen ion (proton) per molecule when it dissociates in water.

5 **a** Substance that contained oxygen **b** Substance containing hydrogen **c** Substance that donates H^+ ion

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ISBN 978-1-92561-015-4